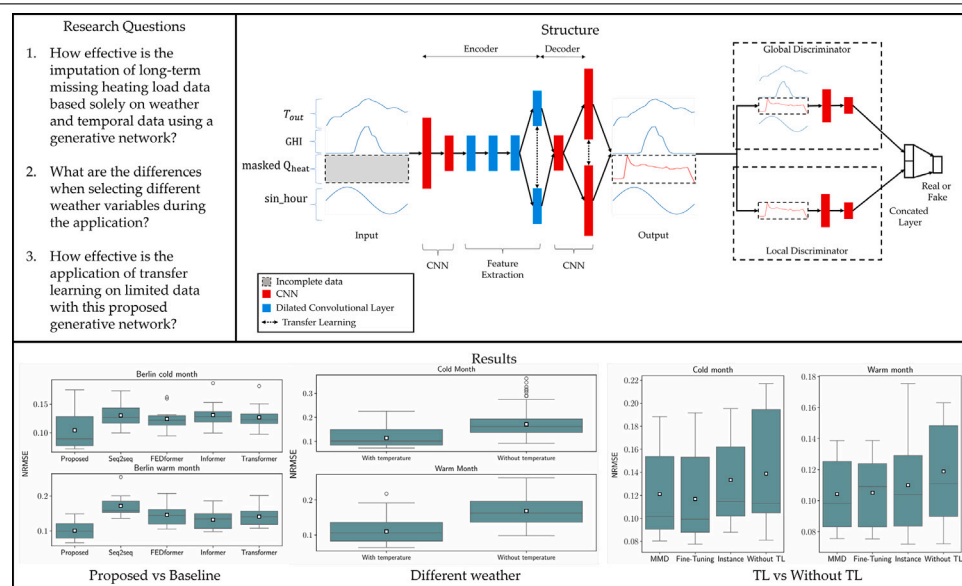


Imputing the long-term missing heating load data using a generative network

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GRAPHICAL ABSTRACT



HIGHLIGHTS

- A novel 2D generative network for imputing long-term missing heat load data.
- Superior accuracy versus state-of-the-art models across diverse seasons.
- Improved interpretability via analysis of key weather variable impacts.
- Effective with limited data through multiple transfer learning strategies.

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ABSTRACT

Accurately filling in missing heating data is essential for ensuring data quality in applications such as energy management optimization and building efficiency analysis. Traditional machine learning methods use historical heating data as an input feature to predict the following missing data. However, when the duration of missing data is long, previous estimated values are inevitably used for further imputation, leading to error accumulation and a growing deviation from true values. To overcome this problem, this paper proposes a generative network that can fill missing data solely based on weather and temporal data, without using previous imputed values for

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Missing data imputation
Transfer learning

further imputation. Our method outperformed the state of the art such as Seq2seq and Transformer, achieving relative normalized root mean square error (NRMSE) reductions of 1.65 % to 41.38 %, 0.30 % to 66.43 %, and 14.84 % to 50.22 % across three different data sources. In addition, with our proposed method, the effect of selecting different weather variables on model performance, and the benefits of transfer learning under limited data were also demonstrated. The relative NRMSE reduction is between 3.88 % to 15.85 % in cold months and from 7.49 % to 12.29 % in warm months when applying transfer learning.

1. Introduction

Ensuring the quality of heating load data is crucial for the effective implementation of optimized energy management systems, particularly against the backdrop of increasing energy demands and the need to achieve sustainable development goals. A common challenge in this context is the imputation of missing data. Heating load data are mainly taken in the form of time series data [1,2], which allows us to apply methods often used for time series data tasks, such as long short-term memory (LSTM), bidirectional LSTM (Bi-LSTM) and Gated Recurrent Unit (GRU) [3]. Currently, ML methods mainly use historical data to impute future missing values via supervised learning approaches such as XGBoost [4], LSTM [5], and Bi-LSTM [6].

However, existing methods typically rely on autoregressive input, that is, previously observed or already imputed load values. A major *challenge* in long-term missing data imputation is that these methods must repeatedly rely on previously imputed values for future predictions. Since imputed values inevitably differ from the ground truth, errors accumulate over time. As the length of the missing period increases, the discrepancy between the imputed and actual values increases, leading to progressively larger deviations.

In order to address the above-mentioned research gap, our objective is to use a generative network to impute the long-term missing values only using the contemporaneous weather and temporal data as the contextual information, thereby avoiding the recursive error propagation in long-term missing data imputation caused by the previous compensation methods. To the best of our knowledge, few studies have been conducted on how to avoid accumulating errors when imputing long-term missing values in heating load data. This network was inspired by a contextual encoder for image inpainting proposed by Iizuka et al. [7], which allows the underlying relationship between unobserved data and the observed data surrounding them to be learned. Historical heating load data, weather data, and temporal data are summarized for every 24 h as multivariate time series data. A pre-trained model based on this network is then trained with these training data, thereby the relationship between heating load data and other variables can be learned. This pre-trained model will impute long-term missing values for every 24 h, based solely on contemporaneous weather and temporal data. This is independent of any prior predicted missing values. This prevents recursive error propagation and maintains fidelity, even over extended periods of missing data.

We compared the performance of our proposed method with other state of the art (SOTA), such as Sequence to Sequence (Seq2seq), Transformer [8], FEDformer [9] and Informer [10]. To improve the physical interpretability of this network, we compared its performance when trained using different weather variables, to evaluate which variables are most effective. Moreover, due to the limited historical data available for most buildings in real-world applications, we also investigated the potential of using a transfer learning method with our network to validate its effectiveness when data volumes are limited. This ensures its suitability for real-world applications.

The main contributions of this paper are described as follows:

1. We reframe long-term missing heating load data imputation as 24 h independent completion tasks, avoiding recursive error accumulation that affects autoregressive methods.
2. We present a context encoder based on Generative Adversarial Networks (GANs) that leverages weather and temporal variables to achieve precise interpolation without requiring any measured heating load data as reference.
3. We evaluate transfer learning strategies (fine-tuning, instance reweighting, MK-MMD) to adapt the model across different buildings, improving generalization with limited training data.
4. We analysed the performance of this network by using different weather variables to enhance physical interpretability.

The rest of this paper is organized as follows. Section 2 introduces previous studies related to impute the missing heating load data using machine learning (ML) and transfer learning approaches to solve heating load data tasks. The following two sections introduce the methodology and the datasets respectively. Section 5 presents and discusses the experimental results. The final section discusses the conclusions and future work.

2. Related works

2.1. Missing data imputation in building energy data

With the development of ML techniques, ML approaches have become more and more popular in solving imputation tasks [11]. A common method is to use supervised learning methods to predict missing values using data from before the missing time range. Li et al. [12] developed a backpropagation neural network model to accurately reconstruct missing air-conditioning power consumption data in a large public shopping mall, achieving relative errors of $\pm 0.8\%$. Lee et al. [4] conducted a comprehensive study for using different techniques and showed, that CART yields the better performance, while XGBoost offers a practical trade-off when computation time is limited. Models that are better suited for processing the time series data, such as LSTM and Bi-LSTM, have also been used to impute the missing data. Hussain et al. [5] proposed an LSTM model combined with a CNN to predict missing values in electricity consumption time series datasets. This approach significantly improved performance compared to using only a CNN or a LSTM, with a relative reduction in RMSE of 48.4 % and 48.8 %, respectively. Xing et al. [6] also used a Bi-LSTM model to obtain the predicted values for building consumption data, demonstrating the superior performance of the proposed method in filling long-term missing compared to traditional statistical-based imputation methods and baseline models.

Apart from supervised learning methods, the unsupervised learning method of using an autoencoder (AE) has also proven highly effective for filling missing values in energy data. For instance, Jeong et al. [13] introduce a Denoising Masked AE that marries the noise-robust latent representations of a denoising AE with the high-mask-ratio capability of a masked AE, enabling accurate reconstruction of missing electric load data with durations ranging from 3 h to 12 h. Miranda et al. [14] showed that AE, when trained offline using complete SCADA datasets, can learn the nonlinear manifold of power-system measurements. When incomplete inputs are fed through the encoder-decoder pathway, missing voltages, power flows and switch status information can be accurately reconstructed. Building on the denoising AE paradigm, Liguori et al. [15] introduce a physics-informed denoising AE that incorporates a basic resistance-capacitance thermal balance equation as a soft constraint within the loss function. This results in a relative reduction in the RMSE when reconstructing heating data (58.6 %), cooling data (16.7 %), and

Nomenclature	
Acronyms	
AE	Autoencoder
Bi-GRU	Bidirectional GRU
Bi-LSTM	Bidirectional LSTM
CNN	Convolutional Neural Network
CVRMSE	Coefficient of Variation of Root Mean Square Error
FC	Fully Connected Layer
GAN	Generative Adversarial Network
GHI	Global Horizontal Irradiance
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MBE	Mean Bias Error
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMD	Maximum Mean Discrepancy
MSE	Mean Squared Error
NRMSE	Normalized Root Mean Square Error (unitless)
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
Seq2seq	Sequence to Sequence
SOTA	State of the Art
TCN	Temporal Convolutional Network
Greek Symbols	
λ	Weight hyperparameter
θ	Model weight
ϕ	Mapping function
Latin Symbols	
X	Multi-variate time series data
j	j th variable in multi-variate time series data
i	Time step i
N	length of multi-variate time series data
w	Rolling window length
$G(\cdot), D(\cdot)$	Generator, Discriminator
t_G, t_D	Training epoch of generator and discriminator
M_G, M_D	Binary mask in generator and discriminator
L_{rec}	Reconstruction error
p_S, p_T	Density of source- and target-domain
w	Ratio of re-scaling source samples
H_k	Reproducing kernel Hilbert space
d_k	Probability distribution
r	Pearson's correlation coefficient

indoor temperature data (14.9%). Other techniques, such as adaptive high-order singular value decomposition techniques [16] and tensor decomposition [17] have been applied to the same problems in other domains, such as tunnel digital twin systems and traffic data, demonstrating how these methods can leverage spatio-temporal correlations to efficiently handle the spatio-temporal missing values. These studies demonstrate the potential of decomposition methods, equally worthy of reference in this study.

Recently, other studies using more advanced ML methods have also been applied to the imputation of long-term missing data in building energy data. In line with the objective of this paper, we conducted a review of related work focusing on two distinct approaches: using the method adapted from image domain and the use of GANs for long-term missing data imputation. Studies presented by Liu and Liu [18] and Fu et al. [19] have also applied the methods to impute missing image data

to missing building energy data. On the other hand, studies on imputing missing data in sensor networks using GANs provide valuable insights for our work. Two GAN-based pipeline studies [20,21] apply generative adversarial training to restore physical sensor fields: [20] proposed a hierarchical (coarse-to-fine) GAN for pipeline pressure recovery emphasizes multi-scale reconstruction to reestablish large-scale structural consistency before refining local details, while [21] used a multi-view framework fuses heterogeneous sensor streams to detect leaks even with incomplete sensors by exploiting cross-sensor coherence.

2.2. Transfer learning

Using data-driven methods for solving the heating load data related tasks often requires a large amount of historical data to train the model. However, as mentioned in Section 1, in real-world cases, there are always conditions of limited data due to lack of monitoring system or time of data accumulation, which makes it difficult to collect a large amount of data to train the model. Transfer learning aims to improve the performance of learners in target domains by transferring knowledge from different, but related, source domains. This makes the transfer learning approach more effective in reducing dependence on large amounts of target domain data when constructing models to solve tasks on target buildings with limited data [22]. According to [23], transfer learning methods can be divided into four main approaches: *instance transfer learning*, *feature representation transfer learning*, *parameter transfer learning* and *relational knowledge transfer learning*.

Several studies have focused on applying transfer learning to tasks related to heating load data. This has also given us an insight into how we can apply a transfer learning approach to our proposed method. For example, Lu et al. [24] used a parameter-based method to forecast the heating load between the source and target domains, achieving a result similar to the target domain by fine-tuning and retraining the transferred parameters. Bai et al. [25] employed data mining techniques such as clustering and classification, investigated the correlation between the features of source building data and the most effective transfer learning hyper-parameter combination, developed a recommendation model to suggest suitable transfer learning hyper-parameter combination based on the specific features of the building heat load data samples, which reduced the prediction error of the target buildings by 0.12 % to 6.64 % compared to the conventional method of empirically determining transfer learning hyper-parameter settings. Another approach using a parameter-based method was proposed by Wang et al. [26], who used a layer transfer model as part of their proposed model for predicting the heating load of a district heating station. The coefficient of variation of the RMSE of this layer transfer model with insufficient training data was reduced by 7.62 %. By segmenting the original heating load data into different segments and extracting the least correlated sequence, Zhou et al. [27] established an adaptive generalized prediction model based on adaptive LSTM and GRU for heating load data forecasting by performing distribution matching over periods. For the application of feature-based transfer learning on heating load forecasting, Li et al. [28] used maximum mean discrepancies (MMD) as a metric to evaluate the similarity between source domain and target domain, in order to identify, whether to adjust the model structure during the transfer learning process, which reduced the RMSE by 12.91 % to 30.63 %. Based on the aforementioned studies, the application of generative networks using similar transfer learning methods to fill in missing values in heating load data is also rationalized, thereby addressing the effects of insufficient data volume.

2.3. Limitations and novelty

However, previous studies each have distinct *limitations*:

- Supervised and unsupervised methods: These approaches have mainly been validated on short gaps of only a few hours. When missing periods are longer, they inevitably rely on previously imputed values to impute further missing data, leading to error accumulation over time and progressively larger deviations from ground truth.
- Image-domain methods: Techniques adapted from image completion can impute entire missing gaps in one step, avoiding recursive errors. However, they often ignore variables such as weather and temporal features that can affect the energy load data, and are trained across multiple buildings. Due to the heterogeneity of building types and the scarcity of large multi-building datasets, such models struggle to generalize.
- GAN-based methods: Existing GAN studies in sensor networks exploit strong spatial correlations among sensors to restore missing data or detect anomalies. Heating load data, however, does not exhibit such spatial patterns: each building responds differently to the same weather conditions depending on its thermal properties, HVAC systems, controls, and occupancy. Moreover, these GAN methods primarily target short gaps and has available measured data as contextual information during the imputation, leaving their effectiveness for extended missing periods unclear, especially when there is no measured data during the target time range.

To overcome these limitations, we propose a novel framework that leverages cross-variable temporal patterns by using a contextual encoder, and segmented imputation to robustly impute long-term missing data in heating load data. Besides, we further explore transfer learning to enable generalization across buildings with limited data. The *novelties* of our method are:

- Temporal feature-based imputation: Our method exploits the strong correlations among weather, temporal, and heating load data, rather than relying on spatial correlations or ignoring other variables, thereby making the network more suitable for imputing the long-term missing data in heating load data and improving the performance.
- Contextual Encoder and Discriminators: The contextual encoder captures cross-variable dependencies between heating load data, weather and temporal variables within the same time range. This enables weather and temporal variables to serve as solid contextual information, providing hints for imputing in missing data. Local and global discriminators ensure imputation align with intrinsic load patterns and their correlation with weather and temporal background.
- Long-term missing periods are divided into 24-hour segments, each imputed independently using only weather and temporal features. The missing heating load data is imputed using only temporal and weather features, avoiding error accumulation in autoregressive methods.
- Transfer learning potential: Applies transfer learning for missing value imputation, enabling accurate results across buildings with limited data—a novel use in generative network-based imputation.

3. Methodology

3.1. Modelling data samples

In this paper, we used the rolling window approach, which is commonly used for resampling time series data, to model the multivariate time series data used for training. Splitting the entire time series data into samples of a fixed length enables the model to be fitted to these segments to capture the relationships between the variables.

Let $\mathbf{X} \in \mathbb{R}^{n \times m}$ be the matrix that represents the multi-variate time series data containing weather data, heating load data and temporal

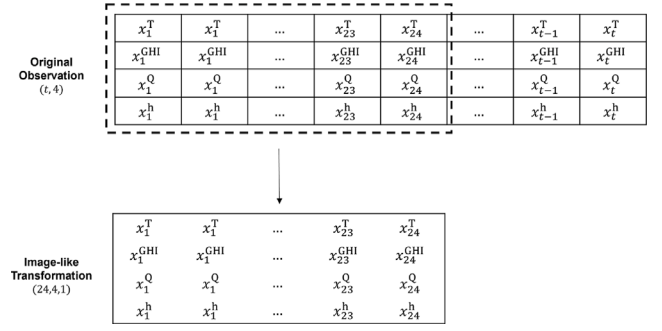


Fig. 1. Structure of the proposed network.

data, where x_{ij} represents the observation of the j th variable at time step i . m is equal to 4, because we contain temperature, GHI, the heating load data and hour of the day. To estimate training samples using a rolling window approach on our multivariate time series data, we define w as the window length and Δt as the step size when estimating the data samples on each dimension simultaneously. In this paper, both w and Δt are 24. The whole time series will be finally partitioned into $(N - \Delta t)/w$ samples.

The outdoor temperature and GHI were selected as weather variables, and the hour as the temporal variable, to impute missing values. This is because these are the weather variables most commonly selected in previous studies for solving heating load data-related tasks [29–32]. Each input sample is then reshaped from its original 1D multivariate time series form into a 2D image-like tensor where the horizontal axis represents time steps and the vertical axis represents different input variables. In this representation, each row corresponds to a different variable over time, effectively treating the variables as the spatial dimension and time as the temporal dimension. This transformation enables the use of 2D convolutional filters to jointly capture temporal dynamics and inter-variable relationships. For clarity, Fig. 1 illustrates this transformation process and resulting tensor shape, where t represents the entire time range covered by the training data. All data from the four variables within the same time range is resampled every 24 h and reshaped from one-dimensional time-series data into two-dimensional image-like data.

In our framework, heating load, weather variables, and temporal features are jointly represented as multi-channel inputs. Heating load data with missing values are concatenated with contemporaneous weather and temporal variables, then reshaped into a two-dimensional format. This design allows the contextual encoder to learn cross-variable dependencies: weather data provide environmental impact of heating load, while temporal features capture usage patterns and operational schedules. Together, these external contextual information guide the network in imputing missing load data more accurately.

3.2. Proposed generative network

Our generative network based on the architecture proposed by Iizuka et al. [7], which was originally intended for the imputation of missing image data. The structure of the proposed network is shown in Fig. 2. This masked data represents the heating load. Above and below the masked heating load data are the weather and temporal data respectively. In the work of Iizuka et al. [7], the contextual encoder was originally developed for image completion tasks, where it reconstructed missing regions by capturing both local textures and global contextual information across spatial dimensions. This concept can be meaningfully extended to the domain of heating load data estimation. Although heating data are inherently temporal rather than spatial, their variation over time exhibits structured patterns that are influenced not only by their own temporal dynamics but also by

external factors such as outdoor temperature, solar irradiance, and time of day. These dependencies introduce inter-variable relationships that resemble the spatial correlations found in image data. Consequently, the ability of the contextual encoder to model local–global relationships makes it a suitable architecture for reconstructing missing segments in multivariate time series, such as heating load data.

In this network, the generator and discriminator engage in a two-player game. The generator strives to produce realistic values that fool the discriminator, which provides adversarial feedback by learning to distinguish generated values from the truth. This drives the generator to produce increasingly realistic reconstructions. Section 3.2.1 and Section 3.2.2 will introduce the generator and discriminator in this network respectively. The training procedure and other setting for build and training the proposed network, including network structure, hyperparameter settings, and comparison models used to verify the effectiveness of this network, is provided in the [Appendices A and B](#).

3.2.1. Generator

The generator in this network is set up as a denoising AE structure. During training, the encoder attempts to reconstruct the heating load data using contextual information about the weather and temporal information. The decoder then recovers the data with complete heating load values. The reconstruction error between the reconstructed data and the original data is used as the loss function to continuously optimize the generator during training, thereby enabling it to achieve increasingly better performance. The training process enables the underlying relationship between weather data, temporal data and heating load data to be captured and built, allowing robust, context-driven imputation without relying on predicted missing values.

As shown in [Fig. 2](#), the generator consists of two main modules: CNN layers and dilated convolutional layers. In our design, each sample is represented as a two-dimensional matrix. One dimension represents the variation of each variable within a specific time range, while the other dimension represents the different variables within this same time range. When multivariate time series data are transformed into 2D representations, a comparable structure emerges. The variables within the specific time range can be treated as one axis, while the input variables, such as weather and temporal variables, can be treated as the other axis. This structured representation enables the network to exploit spatial coherence in a manner analogous to image processing, allowing it to identify meaningful temporal and cross-variable patterns. Treating each variable as a ‘spatial’ dimension and time as the other enables 2D CNNs to learn dependencies across different series and dynamics within each series in a single operation. By summarizing different variables represented as time series data into a 2D multivariate data, the model can apply 2D convolutional filters across both dimensions to capture complex dependencies between multiple sequences. This transformation enables the use of 2D kernel functions to capture complex dependencies between multiple sequences, which has been widely applied in signal data with properties similar to time series data and has achieved significant results [33–35]. 2D CNNs can seamlessly process spectrograms or other image-like embeddings of signals, opening the door to richer feature spaces [35].

Dilated convolutional layers also play a role in making this network, which was adapted for image imputation, effective for filling in missing heating data. These layers can enlarge the receptive field effectively without increasing the number of parameters, which is important for capturing multi-scale contextual information without excessively increasing computational costs or losing fine-grained information [36]. The expanded receptive field enables the network to capture multi-scale temporal patterns and consider contextual information across different variables, such as temporal and weather variables. This enables the network to capture interactions between variables more effectively. As mentioned above, heating load data often exhibits complex temporal structures spanning multiple timescales. These can range from short-term fluctuations due to daily usage patterns, to longer-term variations

driven by weather conditions or seasonal cycles. Dilated convolutions enable the network to integrate information from nearby and distant time steps simultaneously, allowing it to infer missing values based on broader temporal contexts. Furthermore, when applied to the 2D representation of multivariate time series, dilated convolutions facilitate learning cross-variable interactions over extended temporal windows. This helps the network to capture the intricate dependencies between variables such as outdoor temperature, GHI and hour of the day, all of which influence the heating load. Consequently, dilated convolutions increase the ability to produce coherent, contextually relevant estimates of missing heating data, even in scenarios with significant temporal gaps.

3.2.2. Discriminator

In addition to the generator, the discriminator also plays a critical role during the training process of the generative network. While the generator focuses on reconstructing the missing heating load data based on available contextual inputs, the discriminator serves as a learned evaluator that distinguishes between the reconstructed data and the original data. This adversarial interplay between the generator and discriminator forms the foundation of generative adversarial training. Follows the configuration of the generative network proposed by Iizuka et al. [7], in our network, two discriminators are employed during the adversarial training procedure, i.e. global discriminator and local discriminator. The global discriminator assesses whether the filled values are contextually reasonable in the context of surrounding weather and time variables by checking whether the reconstructed heating load data is consistent with the expected multivariate relationship observed in the original data, in order to evaluate, whether the filled data reflect the appropriate heating demand corresponding to specific outdoor temperatures, solar irradiance levels, or time-of-day indicators. In doing so, the global discriminator helps the generator learn to embed domain-specific dependencies and generate heating load patterns that are both statistically and physically meaningful within the broader context. Rather than considering whether the imputed values are plausible in the context of the surrounding weather and temporal variables like the global discriminator, the local discriminator focuses solely on the temporal consistency of the heating data itself. It assesses whether the filled values conform to the fine-grained temporal patterns of the original heating data, such as periodicity, trend continuity and local variability.

3.2.3. Novelty over original context encoder

Our proposed method builds upon the original image-based context encoder while introducing several key innovations tailored for building energy data. First, by adapting a model initially designed for image imputation, we demonstrate the potential of cross-domain transfer, showing how methods from other domain can effectively address challenges in energy data analysis. Unlike the original approach, which primarily focused on handling sporadic missing pixels, our method is designed to impute entire dimensions of missing data, thereby enabling the reconstruction of long-term gaps in building energy records.

Second, while the original context encoder relies on spatial pixel information, we reinterpret the role of contextual information in the energy domain. Specifically, information from auxiliary variables such as weather and temporal features is treated analogously to surrounding pixels in images. Through convolutional neural networks, our method captures inter-variable relationships and leverages them as semantic prompts. This design allows the model to integrate complementary information from multiple data sources, leading to more precise and robust imputation results. This approach is particularly meaningful for building energy data, which exhibits strong temporal patterns such as daily and seasonal cycles and is highly correlated with weather and temporal variables. Leveraging these variables as contextual information enables more accurate and reliable imputation.

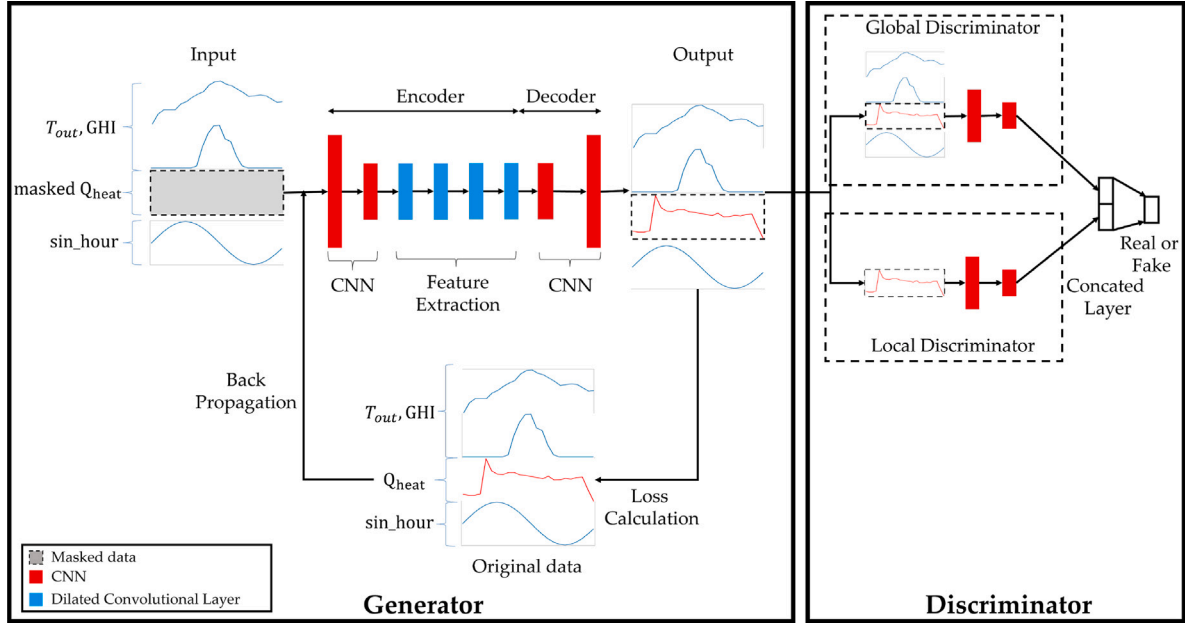


Fig. 2. Structure of the proposed network.

3.3. Transfer learning approach

To apply the transfer learning approach, the building with a large amount of high-quality data is used as the source domain, while buildings with a limited amount of data are used as the target domain to estimate the heating load data. We apply transfer learning exclusively to the generator, pre-training it on source and target domain, while the discriminator remains an auxiliary component, providing adversarial guidance during adaptation without itself being transferred, thereby sharpening the performance of the generator on the target data. Three different approaches in transfer learning, i.e. instance-based transfer learning, fine-tuning and domain adaption using the multiple kernel variant of MMD (MK-MMD) were used. The effectiveness of the samples were then compared with direct training with limited data.

Fine-tuning adjusts pre-trained models to fit the target task, refining parameters based on specific task requirements, which is a straightforward approach that allows a single model to adapt to various tasks with minimal adjustments [37]. In our approach, we freeze the encoder to keep the weights used to capture general features fixed, and only retrain the decoder using data from the target domain. Updating the decoder layers alone tailors the model's output mappings to new data, while preserving robust, low-level feature extractors. This reduces the risk of over-fitting and the computational cost compared to full end-to-end tuning. To illustrate the setup using fine-tuning better, Fig. 3 shows the procedure of using fine-tuning as one of the transfer learning solutions for our network.

Instance-based transfer learning re-weights individual source-domain samples so that the training distribution more closely matches the target distribution, under the covariate-shift assumption that only the input density changes between domains [38]. Eq. (1) shows the mathematical expression of instance-based transfer learning, where x_i is the i th source-domain input, $G_\theta(\cdot)$ is the function of generator parameterized by θ , and $\|x_i - G_\theta(x_i)\|^2$ is the reconstruction error. The weight $w(x_i)$ re-scales each sample's contribution by the ratio of its target-domain density $p_T(x_i)$ to its source-domain density $p_S(x_i)$, thereby emphasizing those source examples most representative of the target distribution.

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n w(x_i) \|x_i - G_\theta(x_i)\|^2, \quad w(x_i) = \frac{p_T(x_i)}{p_S(x_i)}, \quad (1)$$

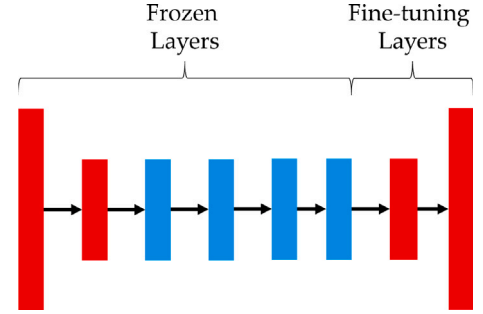


Fig. 3. Illustration of fine-tuning approach.

Whereas instance-based transfer learning tackles covariate shift by re-weighting source samples to mirror the target distribution, domain adaptation methods instead focus on learning domain-invariant feature representations to directly align source and target distributions [39]. The third approach to transfer learning is to use a domain adaptation-based approach involving a multiple kernel variant of maximum mean discrepancy (MK-MMD) [40]. According to Li et al. [41], MK-MMD is an UDA algorithm designed to improve the performance of cross-domain learning by minimizing the differences between the source and target domains. This algorithm utilizes a distance metric called MMD to quantify the distance between two distributions, allowing the mapping of the source and target domains into a feature space and computing their distances in that space. MK-MMD algorithm quantifies the differences between the source and target domains by maximizing the MMD, achieving domain adaptation.

Assuming H_k is the reproducing kernel Hilbert space endowed with a characteristic kernel k , X_s and X_t are the sample sets from source domain and target domain respectively, and ϕ_k is the mapping function which can map the original data into H_k , the MK-MMD $d_k(X_s, X_t)$ between probability distributions source domain X_s and target domain X_t is defined as the RKHS distance between the mean embeddings of X_s and X_t . The MK-MMD formulation can be defined as in Eq. (2), the goal of MK-MMD is to maximize this formula.

$$d_k^2(X_s, X_t) = \|\mathbb{E}_p[\phi(X_s)] - \mathbb{E}_q[\phi(X_t)]\|_{H_k}^2 \quad (2)$$

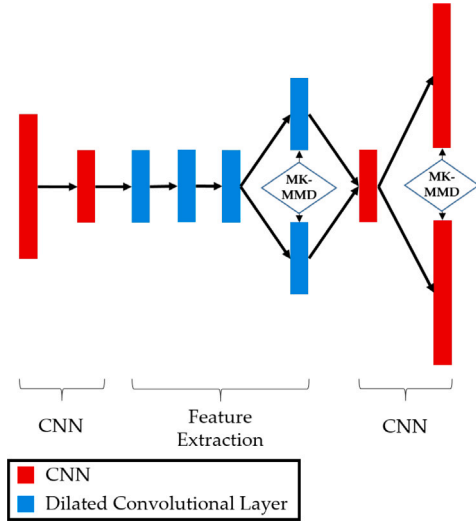


Fig. 4. Illustration of domain adaption approach.

The illustration of applying MK-MMD for transfer learning is shown in Fig. 4, where we adapt the last layer in the feature extraction part and the output layer of the network with MK-MMD. The last layer in the feature extraction part and the output layer of the network are tailored to fit specific tasks, hence they are not transferable with simple fine-tuning approach and should be adapted with MK-MMD.

3.4. Evaluation metric

In order to evaluate the error between estimated data and original data between our method and baseline methods, we selected three different metrics, NRMSE, Mean Absolute Error (MAE) and Mean Bias Error (MBE) to evaluate the difference between the estimated data and the original data from different views. NRMSE normalizes RMSE to make them comparable across different scales. MAE measures the average absolute error in the original data, providing a clear indication of typical deviation. MBE captures systematic bias, indicating whether the model consistently over- or underestimates true values. Eq. (3) to Eq. (5) show how these three metrics are calculated.

$$\text{NRMSE} = \frac{1}{x_{\max} - x_{\min}} \cdot \sqrt{\frac{\sum_{i=1}^n (x_i - \hat{x}_i)^2}{n}} \quad (3)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{x}_i| \quad (4)$$

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i) \quad (5)$$

Where $x_{\max}$ and $x_{\min}$ are the maximal and minimal values in the heat load data X , n is the number of data points in corresponding time period, x_i and $\hat{x}_i$ are the real value and estimated value respectively.

Another metric introduced in this section is for the comparison between different weather variables and heating load data, which was used as a basis for selecting different weather variables to validate the performance of the estimation result. The correlation was determined using the Pearson's correlation coefficient, which measures the linear relationship between two sets of data. Eq. (6) shows how the Pearson's correlation coefficient is calculated.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (6)$$

Two statistical analyses were conducted to provide a rigorous evaluation of the reconstruction performance. First, we compute the

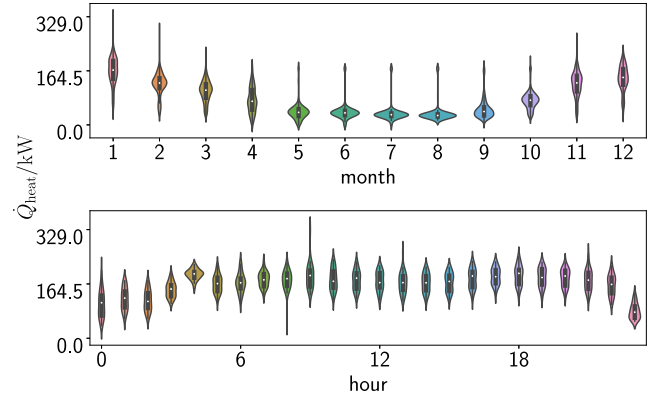


Fig. 5. Distribution of the heat load data from residential buildings in Berlin.

95% confidence intervals for the reconstruction error metrics, including NRMSE, MAE, and MBE. The confidence interval is given as in Eq. (7):

$$CI = \bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \quad (7)$$

where $\bar{x}$ is the mean of the reconstruction error, s is the standard deviation, n is the number of samples, and $z_{\alpha/2}$ is the critical value from the standard normal distribution for the desired confidence level.

Second, we perform a paired nonparametric test using the Wilcoxon signed rank test to evaluate whether the improvements of the proposed method over baselines are statistically significant. This test statistic is defined as in Eq. (8):

$$W = \sum_{i=1}^n R_i \cdot \text{sgn}(d_i) \quad (8)$$

where $d_i = x_i - y_i$ is the difference between paired errors from two methods, R_i is the rank of $|d_i|$ (excluding zero differences), and $\text{sgn}(d_i)$ is the sign function. The corresponding p -value is then obtained by comparing W to its sampling distribution under the null hypothesis.

4. Data foundation

4.1. Heating load data sources

In order to compare the performance of our proposed method with other baseline models, we have selected datasets from three different data sources to validate the performance of the models. Both real-world data sources (Sections 4.1.1, 4.1.3) and simulated data source (Section 4.1.2) are used.

4.1.1. Heating load data from the residential buildings in Berlin

The first data source we have selected is the heating load data from different residential buildings in Berlin. The dataset contains heating load data measured from smart meter data by Green Fusion GmbH from 2023-10-10 to 2025-05-05 for over 100 residential buildings in Berlin. This dataset was also used to explore, whether the selection of stronger correlated weather data can improve the accuracy of estimating the heating load data, and also to apply transfer learning between different residential buildings. Fig. 5 shows the distribution of heating load data over hours and months. From the demonstration, we can find that the heating load data in this dataset shows the difference between different months and a significant day-night pattern.

4.1.2. When2Heat heating profiles

The second data source for evaluating the performance of our proposed method is called When2Heat Heating Profiles. This dataset comprises national time series to represent building heat pumps in

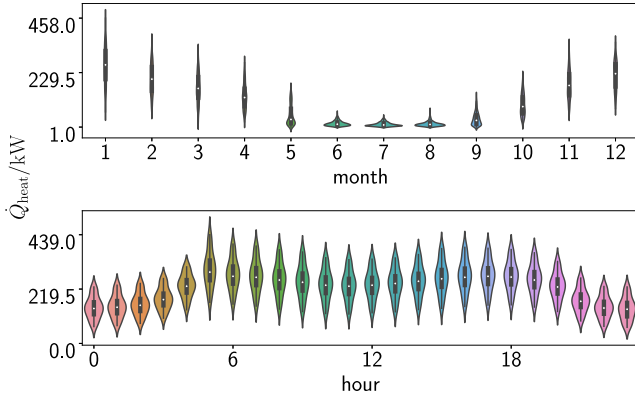


Fig. 6. Distribution of the heating load data from residential buildings in Berlin.

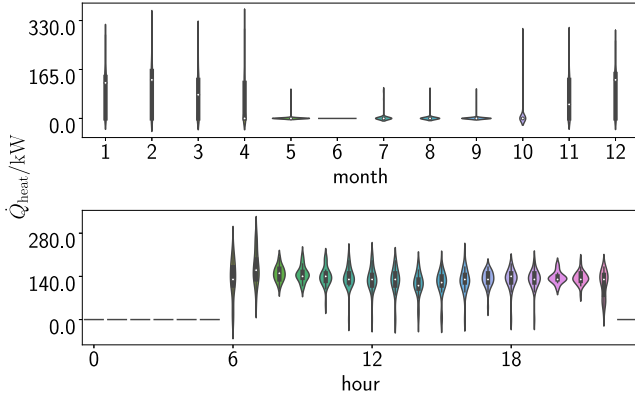


Fig. 7. Distribution of the heating load data from Niš dataset.

power system models. This data source explicitly refers to space and water heating, and contains hourly heat load data from single-family and multi-family houses, as well as commercial buildings, in 28 European countries from 2008 to 2022. [42]. In this paper, we have selected the heating load data of commercial buildings and single-family houses through space heating from the three German-speaking countries, Austria, Switzerland and Germany. Fig. 6 shows the distribution of heating load data over hours and months. Same as the dataset described in Section 4.1.1, heating load data from this data source also shows significant day-night and seasonal patterns.

4.1.3. Niš dataset

The last dataset we used in this paper is the heating load data from the existing District Heating System (DHS) infrastructure, the Faculty of Mechanical Engineering in Niš. Heating load data are based on data acquired from a SCADA system. Fig. 7 shows the distribution of this dataset. As can be seen in Fig. 7, although it also shows a day-night pattern, unlike the datasets described in Sections 4.1.1 and 4.1.2, during the low heating load profile, the heating load data is stable around 0. In warm months, the heating load is close to 0 at all times. For this reason, we only used data from cold months for training in this dataset.

4.2. Weather dataset

The weather data used for Berlin and Niš heating load data are provided by OpenWeather [43]. The weather data from Open Weather is estimated by their own weather model, which uses a combination of variable numerical weather prediction, weather station, weather radar and satellite data. Weather variables were obtained from the

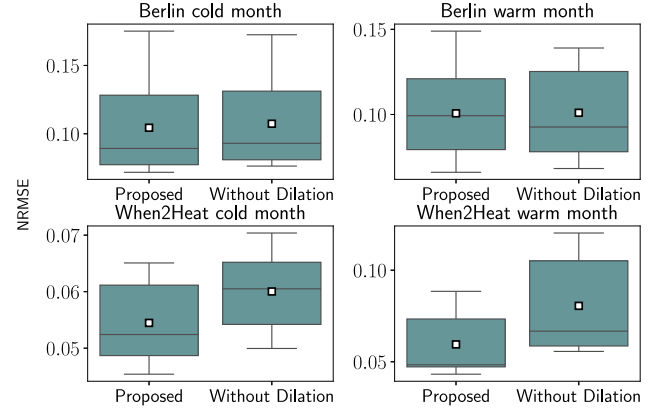


Fig. 8. Comparison of NRMSE with and without dilated convolutional layers for Berlin residential dataset and When2Heat dataset during cold and warm months.

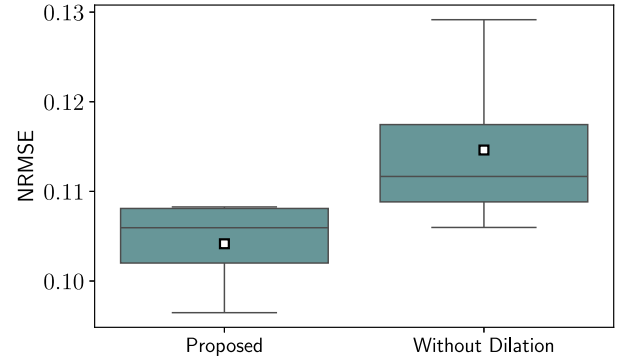


Fig. 9. Comparison of NRMSE with and without dilated convolutional layers for Niš dataset.

OpenWeather historical (hindcast/reanalysis) service, which combines station observations and numerical weather prediction outputs to provide retrospective estimates. These data represent reanalysis rather than operational forecasts and were retrieved with a latency of several hours to one day. It contains several weather variables, such as outdoor temperature, GH1, apparent temperature and dew point temperature, which makes it possible to explore the impact of different weather variables on the estimation.

4.3. Data split and leakage guard

As mentioned in Section 3.1, for splitting the training and testing data samples, we construct fixed-length fixed-length 24h samples with stride 24h. To prevent data leakage, training data and test data are strictly separated in time with no overlapping periods. The selected time range for training data and test data for all datasets are shown in Table 1. The training and validation data used during the training process originate from the same time period. After partitioning the data into distinct samples based on 24-hour intervals, the entire samples are subjected to random shuffling. Then, 80% of the samples from all the training samples are selected as the training data and the remaining 20% as the validation data. For building-level segmentation, we trained a distinct model, either from scratch or through transfer learning between different buildings. This design ensures that each model focuses exclusively on a single building, so variations in heating load patterns across buildings do not bias performance, addressing the limitations discussed in Section 2.3.

Table 1

Overview of training and test data ranges for each dataset.

Dataset	Training time range	Test time range (Cold Months)	Test time range (Warm Months)
Berlin residential building datasets	'2023-11-01 00:00:00' – '2024-12-05 23:00:00'	'2024-12-06 00:00:00' – '2025-01-05 23:00:00'	'2025-04-01 00:00:00' – '2025-05-01 23:00:00'
When2Heat profiles	'2016-01-01 00:00:00' – '2017-12-31 23:00:00'	'2018-01-01 00:00:00' – '2018-04-30 23:00:00'	'2018-06-01 00:00:00' – '2018-09-30 23:00:00'
Niš datasets	'2022-10-03 00:00:00' – '2023-03-31 00:00:00'	'2023-12-01 00:00:00' – '2023-12-31 00:00:00'	–

5. Results

5.1. Ablation study

To investigate how the application of dilated convolutional layers affects the performance of this generative network, we first conducted an ablation study by removing them from the network and compared the results obtained with and without using dilated convolutional layers in our proposed network on the three data sources that have been introduced from Sections 4.1.1 to 4.1.3. Figs. 8 and 9 illustrate the performance comparison with and without dilated convolutional layers in our proposed network. The white cube in this demonstration is the average NRMSE for all buildings. As described in Section 4.1.3, the Niš dataset does not show the difference between cold and warm months because it only contains data from the former.

The results suggest that adding several dilated convolutional layers improves model performance, although this varies depending on the dataset. Adding dilated convolutional layers significantly improved the performance of the models for the When2Heat and XAI4Heat datasets, achieving relative NRMSE reductions of 9.266 % (When2Heat dataset during cold months), 26.128 % (When2Heat dataset during warm months) and 9.125 % (XAI4Heat dataset). However, for the Berlin residential dataset, adding dilated convolutional layers did not significantly improve performance, achieving relative decreases in NRMSE of only 2.740 % (cold months) and 0.316 % (warm months). supported by a visual comparison of the original and imputed data, as shown in Figs. 10 and 11. Both figures show that, using dilated convolution layers achieves better results when reconstructing missing data, regardless of the deviation of the data or the effectiveness of restoring the original data sequence pattern, compared to not using them in the network. Results from Figs. 8 to 11 further illustrated the meaning of using the dilated convolutional layers as described in Section 3.2, the extended receptive field can effectively capture a wider range of contextual information, making it easier to identify interactions between variables. This improves the mapping between heating load data and other variables, enhancing the network's performance.

5.2. Comparison between proposed method and baseline models

In this section, we evaluate the performance of our proposed method against several baseline approaches to demonstrate the effectiveness of the novel components introduced in this work. As described in Section 2.3, the baseline methods are grouped into three categories: (i) autoregressive methods, which use previously observed or imputed load values for prediction, (ii) non-autoregressive methods, which rely on contemporaneous inputs without recursive estimation, and (iii) methods that use heating load data from different buildings as input. These three comparisons collectively highlight the advantages of our method in handling long-term missing data, leveraging cross-variable correlations for imputing without using previously imputed data, and thereby mitigating error accumulation. In addition to compare the errors between original and imputed data, the training time and memory usage between different models are also compared. All experiments were conducted with an AMD Ryzen 9 7950X 16-core processor (4.50 GHz) and 64 GB of RAM.

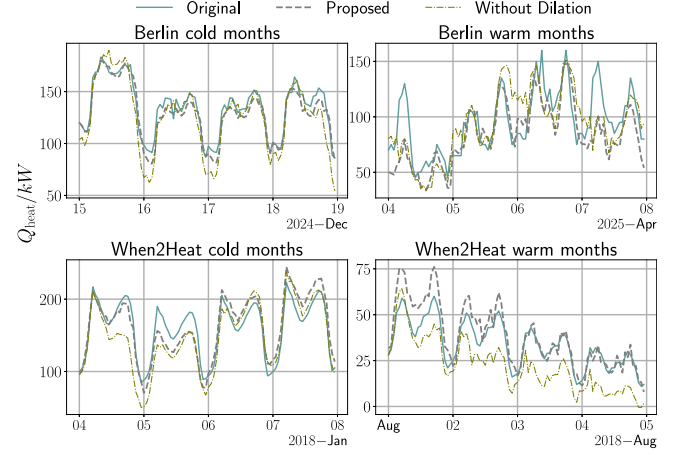


Fig. 10. Comparison of original and filled data with and without dilated convolutional layers for Berlin residential dataset and When2Heat dataset during cold and warm months.

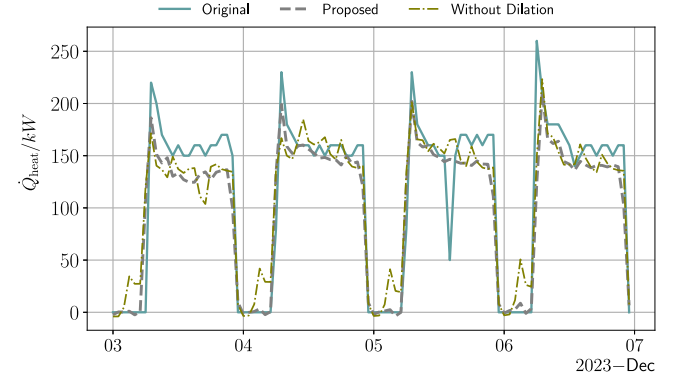


Fig. 11. Comparison of original and filled data with and without dilated convolutional layers for Berlin residential dataset and When2Heat dataset during cold and warm months.

5.2.1. Comparison with autoregressive methods

The first comparison is conducted with autoregressive methods, which rely on previously imputed values to fill subsequent missing data. This evaluation highlights the key novelty of our proposed method: achieving superior performance by avoiding the error accumulation inherent in autoregressive approaches. The selected autoregressive baselines include Seq2Seq, FEDFormer, Informer, and Transformer. Specifically, imputed data from the previous 24h were used to predict the missing data next 24h as target values. In this setup, each set of predicted values was subsequently fed back into the model as input for the next prediction, allowing iterative reconstruction of long-term missing heating load data until all missing values were filled, simulating real-world application scenarios. For all the methods in this comparison, multivariate data containing weather and temporal data, as well as heating load data from the same time range, were used as inputs.

All three introduced datasets were used to compare the performance between our proposed method and other baseline models. Figs.

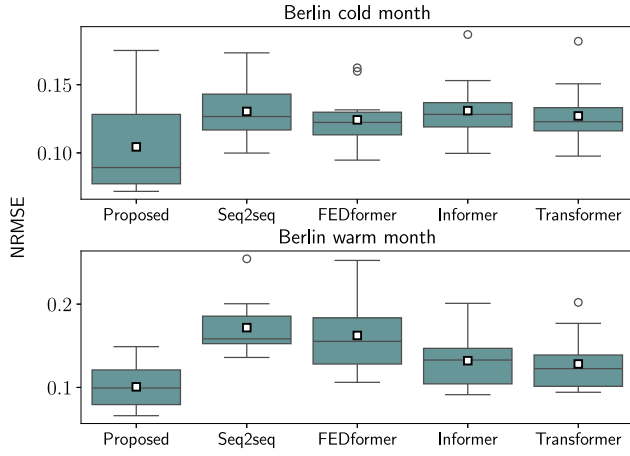


Fig. 12. Comparison of NRMSE on Berlin residential building dataset.

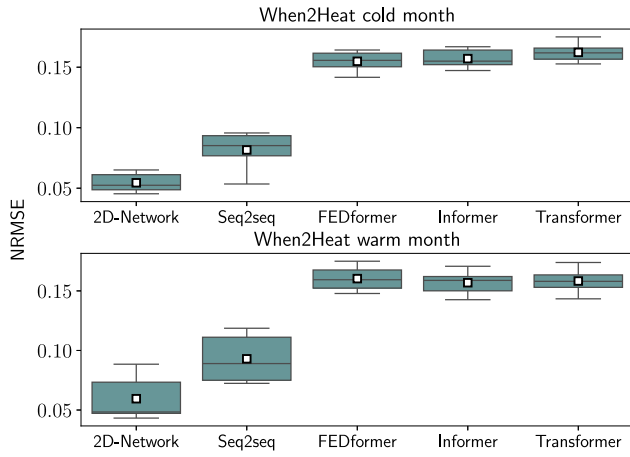


Fig. 13. Comparison of NRMSE on 'When2Heat' dataset.

Table 2

Comparison of NRMSE in % for different methods and datasets.

Dataset	season	Seq2seq	FEDformer	Informer	Transformer
Berlin	cold months	19.94	15.96	20.28	17.86
	warm months	41.38	31.09	23.77	28.63
When2Heat	cold months	33.25	64.83	65.32	66.43
	warm months	36.01	62.89	62.07	62.42
Niš dataset	–	24.05	50.22	47.60	47.13

12 to 14 show the comparison of NRMSE between different methods on the dataset mentioned in Section 4.1.1 to Section 4.1.3 respectively. The white cube is the average NRMSE for all buildings. The demonstration from both figures showed that different methods exhibit dataset-specific strengths, yet none surpassed our proposed method. For example, Seq2Seq preform well on the When2Heat and Niš datasets, whereas FEDformer, Informer, and Transformer achieve better performance on the Berlin residential dataset. Crucially, despite these relative advantages, our model consistently outperforms every baseline across all three datasets, demonstrating both superior effectiveness and robust generalization. To better compare the performance of different methods on different datasets, Table 2 shows the percentage reduction in NRMSE of our proposed method compared to other autoregressive baselines. The comparison of other two metrics, MAE and MBE, are shown in Appendix C.

To specifically compare the differences, a visual comparison of the original and estimated data was then carried out. Figs. 15 and 16 show

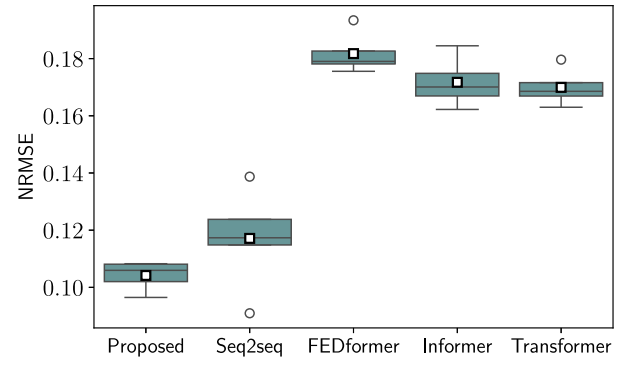


Fig. 14. Comparison of NRMSE on Niš dataset.

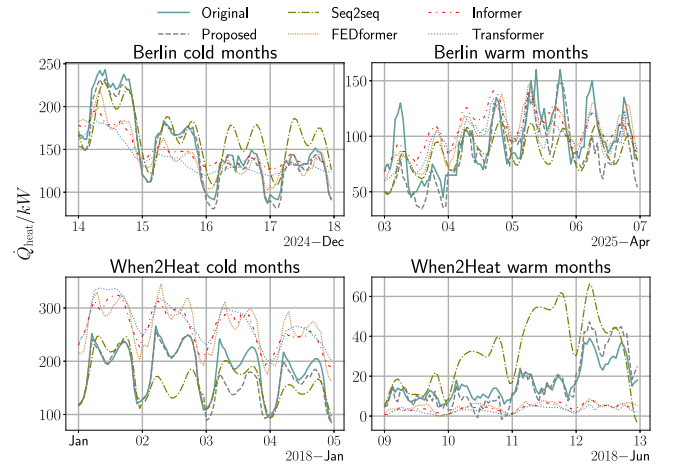


Fig. 15. Comparison of original and estimated data on Berlin residential building dataset.

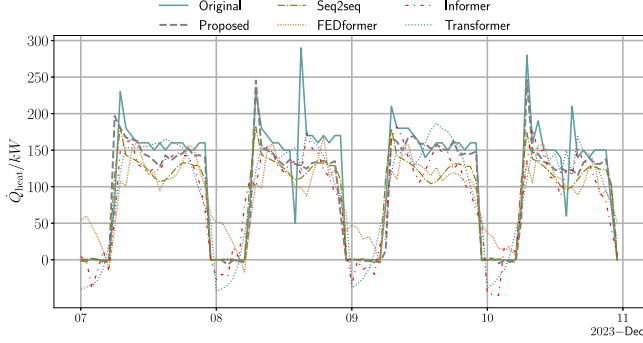
the comparison of the results using different baseline models for all three datasets. The comparison from the two figures further illustrates how effective our proposed method is compared to other baseline methods. It accurately reflects the general patterns and numerical values of the original heating load data across different datasets and different seasons, while other baseline methods encountered different problems when reconstructing data from different seasons in different datasets. For example, when the long-term missing data in the Berlin residential dataset from the cold months was reconstructed, other baseline methods were unable to accurately reconstruct the data when a significant trend change occurred in the heating load data. This is particularly evident in the results obtained using Seq2seq, Informer and Transformer. When the missing heating data for the first two days was reconstructed using the reconstruction graph, which showed no significant trend changes, these three methods performed similarly to our proposed method. However, when the load curve changed suddenly, these three methods failed to reconstruct the trend changes. In these cases, large errors degrade not only immediate predictions of the missing values, but also cascade forward, affecting the estimation of subsequent missing values since each subsequent estimation depends on the previous one. The impact of prediction errors can also be seen in the imputation of missing values for the When2Heat dataset from both cold and warm months. Due to the significant offset between the initial estimation and the original data, this offset persisted in subsequent predictions of missing values, thereby affecting the results.

The comparison also showed the same difference in Niš dataset, as shown in Fig. 16, which the heating load profile differs from that of the other two datasets. The differences between our proposed method and other baseline methods are primarily evident in the reconstruction

Table 3

Comparison of the confidence intervals for NRMSE with autoregressive baselines.

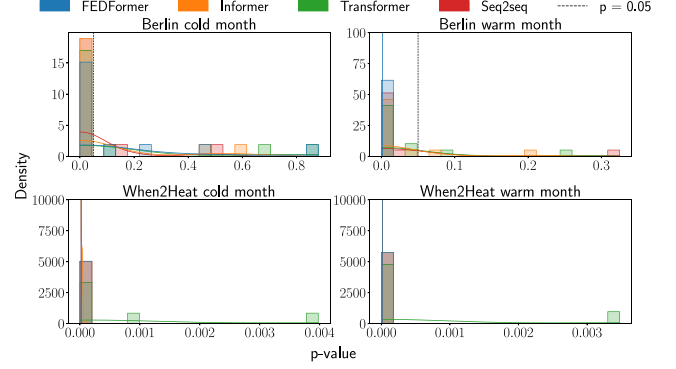
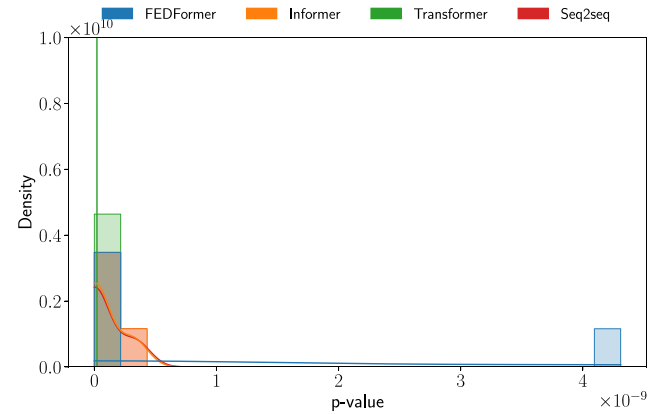
Scenario	Proposed	Seq2seq	FEDformer	Informer	Transformer
Berlin cold month	[9.35 ± 3.47, 11.62 ± 3.30]	[12.29 ± 2.02, 13.77 ± 2.33]	[12.21 ± 2.95, 16.07 ± 2.81]	[13.43 ± 2.50, 17.86 ± 1.87]	[14.27 ± 2.59, 17.52 ± 2.26]
Berlin warm month	[9.15 ± 2.5, 11.30 ± 2.42]	[16.14 ± 3.13, 18.22 ± 3.32]	[14.35 ± 4.48, 16.1 ± 4.74]	[12.29 ± 3.15, 14.15 ± 2.26]	[11.85 ± 3.23, 13.77 ± 3.58]
When2Heat cold month	[5.46 ± 0.74, 5.80 ± 0.77]	[7.90 ± 1.55, 8.42 ± 1.63]	[15.32 ± 0.85, 15.65 ± 0.89]	[15.55 ± 0.80, 15.87 ± 0.82]	[16.05 ± 0.80, 16.40 ± 0.82]
When2Heat warm month	[5.89 ± 1.87, 6.59 ± 2.11]	[8.89 ± 2.05, 9.71 ± 2.10]	[15.29 ± 0.30, 16.77 ± 1.07]	[15.83 ± 1.02, 17.31 ± 1.06]	[15.96 ± 0.96, 17.42 ± 0.99]
Niś cold month	[9.48 ± 0.42, 11.40 ± 0.75]	[10.56 ± 1.34, 13.88 ± 1.21]	[15.04 ± 1.02, 18.20 ± 1.14]	[14.17 ± 1.09, 17.52 ± 1.09]	[14.00 ± 0.69, 17.16 ± 0.69]

**Fig. 16.** Comparison of original and estimated data on Niś dataset.

of low heating load profiles. Our method also successfully reflects the smooth pattern close to 0 for the low heating load profile. However, other methods failed to reconstruct the stable pattern of the low heating load profile. Seq2seq filled the missing values in the low heating load profile with values significantly higher than 0, while the results from the other three methods became negative. When the estimated missing data values are used to predict missing values in subsequent time periods, the results of the subsequent missing value estimation show same result as the previous predictions, i.e., they do not conform to the original data pattern, with significantly greater values than 0 or negative values during low load profiles. This further illustrates the problem of error cascading forward when using previously predicted missing values to impute further missing values via supervised learning methods. The results of imputing the long-term missing values using the method we proposed on three different data sources are shown in [Appendix E](#).

The next evaluation quantifies performance using statistical analysis, reporting 95 % confidence intervals for each building and metric, and conducting paired Wilcoxon signed-rank tests to assess whether the proposed method's improvements over each baseline are statistically significant. Given the large number of datasets across multiple sources, [Table 3](#) summarizes per-building results by aggregating the confidence intervals for each metric, rather than presenting individual building outcomes. For each metric, we report the mean values of the upper and lower bounds of the 95 % confidence intervals as well as their variance, reflecting the variability and reliability of the results. Following testing across diverse datasets spanning different seasons, the results demonstrate that our proposed method exhibits lower upper and lower bounds of the confidence intervals for all three error metrics compared to other baseline methods, whilst also displaying reduced variance. These results conclusively demonstrate that our proposed method delivers superior performance and enhanced robustness.

In addition, we used the Wilcoxon signed rank test to verify whether the performance improvement of the proposed method over the baseline model was statistically significant. The p -value distributions across different datasets from various data sources and across different seasons are presented in [Figs. 17 and 18](#). The results indicate that only on datasets sourced from Berlin residential buildings did a small number of datasets exhibit p -values exceeding 0.05, suggesting that the proposed method's improvements are not statistically significant on these particular datasets.

**Fig. 17.** Statistical significance of paired comparisons (Wilcoxon test) between our proposed method and autoregressive baselines on Berlin residential building and When2Heat Profiles.**Fig. 18.** Statistical significance of paired comparisons (Wilcoxon test) between our proposed method and autoregressive baselines on Niś datasets.

The average training time, memory usage and model capacity of each model on all datasets is shown in [Table 4](#). The training times for FEDformer, Informer, and Transformer are significantly longer than our proposed method, with little difference in memory usage and model capacity. Moreover, as noted above, their performance falls considerably short of ours. Due to its simple model structure, training time is much shorter and model capacity of Seq2seq is much smaller than our proposed method. However, although Seq2seq's training time is shorter, its performance is also poorer, and the memory usage is also similar to our proposed method.

5.2.2. Comparison with other non-autoregressive methods with weather-only input

To further evaluate the effectiveness of our proposed method for long-term missing data imputation, we then compared it against non-autoregressive baselines, including CNN- and TCN-based masked autoencoders with the same depth and bottleneck size as our network. Same as in our proposed method, these baselines reconstruct each 24 h segment in a single step without using previous imputed data, providing benchmarks with the same non-autoregressive mechanism as our proposed method.

Table 4
Training time different methods.

Method	Training time [s]	Memory usage [MiB]	Model capacity [MiB]
Proposed method	214.98	706.83	11.38
Seq2seq	157.30	624.92	1.27
FEDformer	1017.86	632.26	10.31
Informer	505.84	567.94	10.28
Transformer	571.21	563.39	10.27

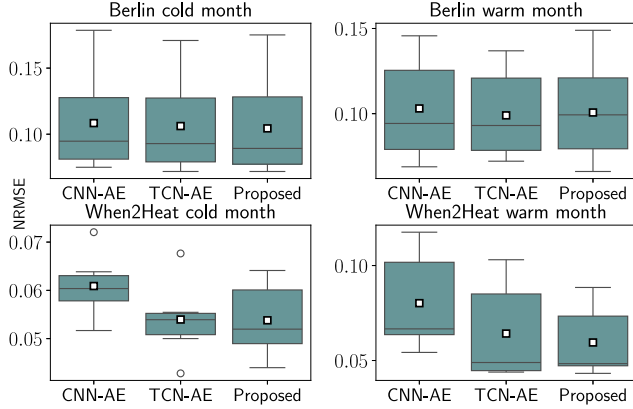


Fig. 19. Comparison of original and estimated data on Berlin residential building dataset.

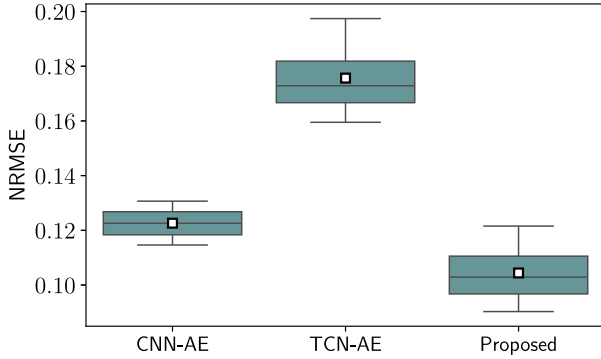


Fig. 20. Comparison of original and estimated data on Niš dataset.

Figs. 19 to 20 show the comparison of NRMSE between CNN masked-AE, TCN masked-AE and our proposed method. The white cube is the average NRMSE for all buildings. The performance of TCN masked AE and CNN masked AE differs across datasets from various data sources. TCN masked AE demonstrates superior performance on datasets sourced from When2Heat Profiles, whereas CNN masked AE performs better on the datasets from Niš data source. Our proposed method, although performs similarly to these two approaches on Berlin residential heating load data, it outperforms both baseline methods on the other two datasets. Table 5 shows the percentage reduction in NRMSE of our proposed method compared to other non-autoregressive baselines with weather-only inputs. This consistent superior performance across datasets with different data patterns further demonstrates the superiority of our proposed model. The comparison of other two metrics, MAE and MBE, are shown in Appendix D.

Same as the evaluation in Section 5.2.1 with autoregressive methods, we also reported 95 % confidence intervals per-building of each metric and conducted paired Wilcoxon signed-rank tests to determine whether the improvement of the performance is significant on statistical meaning. Table 3 summarizes per-building results by aggregating

Table 5
Comparison of NRMSE in % for non-autoregressive baselines with weather-only inputs.

Dataset	Season	CNN-AE	TCN-AE
Berlin	cold months	2.30	-1.70
	warm months	3.69	1.65
When2Heat	cold months	25.81	7.32
	warm months	11.66	0.30
Niš dataset	cold months	14.84	40.56

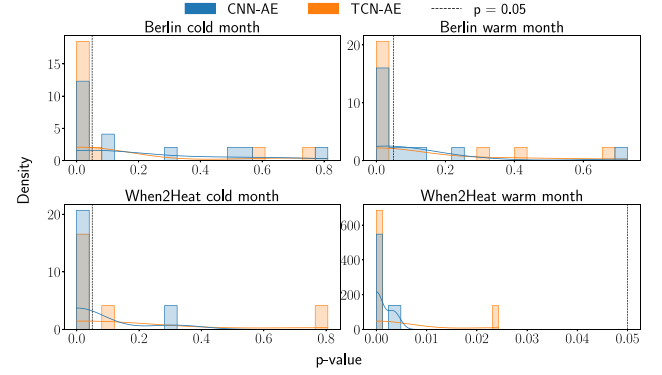


Fig. 21. Statistical significance of paired comparisons (Wilcoxon test) between our proposed method and non-autoregressive baselines on Berlin residential building and When2Heat Profiles.

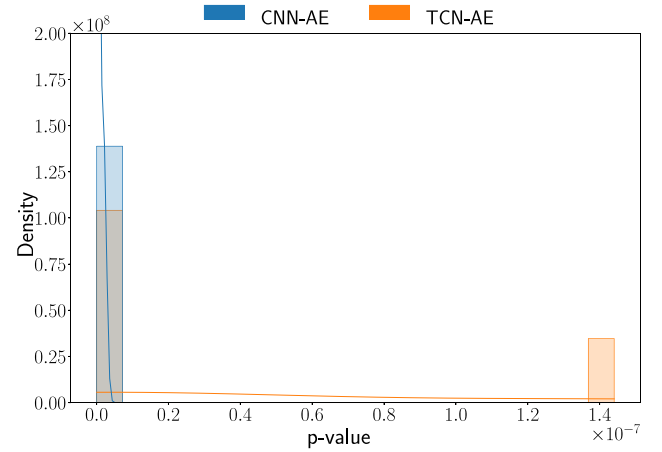


Fig. 22. Statistical significance of paired comparisons (Wilcoxon test) between our proposed method and non-autoregressive baselines on Niš residential datasets.

the confidence intervals for each metric, while Figs. 21 and 22 show the paired test between our proposed method and other non-autoregressive baselines. The results indicate that the confidence intervals for NRMSE, MAE and MBE show minimal variation across different seasons for the two data sources excluding Niš cold month. However, the p-values obtained from paired tests reveal that the majority of datasets yield p-values below 0.05 upon validation. This demonstrates that the improvements achieved by the proposed method are statistically significant (see Table 6).

Table 7 reports average training time, memory usage and model capacity for the compared non-autoregressive methods. Our proposed method requires substantially more training time and has a larger model capacity than the CNN/TCN masked AEs, however, this additional model capacity and training cost yields consistently better imputation accuracy, indicating a trade-off between computational

Table 6

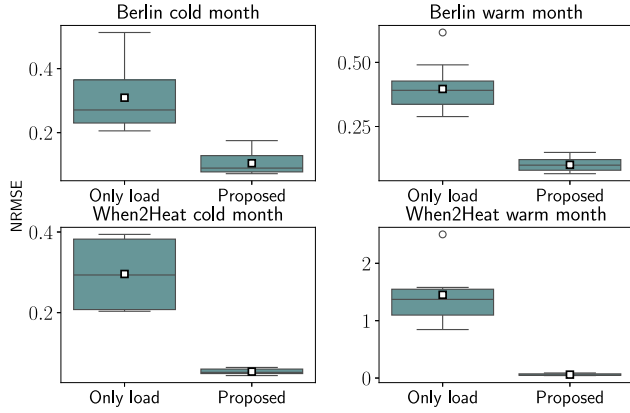
Comparison of the confidence intervals for NRMSE with non-autoregressive baselines.

Scenario	Proposed	CNN-AE	TCN-AE
Berlin cold month	$[9.35 \pm 3.47, 11.62 \pm 3.30]$	$[9.77 \pm 3.54, 11.95 \pm 3.36]$	$[9.5 \pm 3.49, 11.77 \pm 3.35]$
Berlin warm month	$[9.15 \pm 2.5, 11.30 \pm 2.42]$	$[9.29 \pm 2.64, 11.45 \pm 2.59]$	$[8.89 \pm 2.38, 11.04 \pm 2.26]$
When2Heat cold month	$[5.46 \pm 0.74, 5.80 \pm 0.77]$	$[5.89 \pm 2.26, 6.3 \pm 0.72]$	$[5.23 \pm 0.79, 5.56 \pm 0.84]$
When2Heat warm month	$[5.89 \pm 1.87, 6.59 \pm 2.11]$	$[7.51 \pm 2.53, 8.54 \pm 3.00]$	$[6.14 \pm 2.67, 6.7 \pm 2.87]$
Niš cold month	$[9.48 \pm 0.42, 11.40 \pm 0.75]$	$[11.42 \pm 0.61, 13.07 \pm 0.77]$	$[16.6 \pm 1.46, 18.52 \pm 1.71]$

Table 7

Training time different non-autoregressive methods.

Method	Training time [s]	Memory usage [MiB]	Model capacity [MiB]
Proposed method	214.98	706.83	11.38
CNN masked AE	28.53	503.90	1.07
TCN masked AE	48.31	681.77	3.88

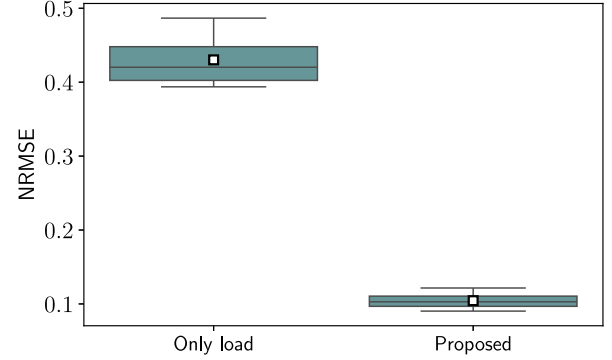
**Fig. 23.** Comparison of original and estimated data on Berlin residential building dataset.

expense and reconstruction performance. Furthermore, memory usage is consistent across models, demonstrating that our method delivers superior accuracy without requiring significantly more memory than the baselines.

5.2.3. Comparison of with and without using weather variables as input

Another comparison was conducted to support the novelty of our proposed method, using different input features. As summarized in Sections 2.1 and 2.3, previous studies often aggregated sensor data from multiple sensors as model inputs when imputing sensor data in the same time series format. However, heating load data from different buildings lacks the strong spatial correlation present between sensors within the same measurement system. Therefore, using data from other buildings to fill long-term missing data is unsuitable for our application. The results comparing different model inputs in this section further demonstrate this.

Figs. 23 to 24 show the comparison of NRMSE between only using heating load data and using input features in our proposed method. The white cube is the average NRMSE for all buildings. The results indicate that using heating load data from other buildings as input for missing value imputation yields significantly poorer performance than employing weather and temporal data. This further demonstrates that weather and temporal data are better suited to imputing missing values in building energy data.

**Fig. 24.** Comparison of original and estimated data on Niš dataset.**Table 8**

Different combined weather variables for estimating heat load data.

Feature	Label
Outdoor temperature	T_{outdoor}
Global Horizontal Irradiance	GHI
Dew point	T_{dew}
Apparent temperature	T_{ap}
Relative humidity	φ
Windspeed	v_{wind}
Cloudcover	C_{cloud}
Precipitation	P

5.3. Comparison between different selection of weather data

Having verified the superior performance of the proposed model, we further investigate its robustness by analysing how variations in physical conditions affect its ability to fill missing heating load data, thereby evaluating the adaptability of the model to diverse environmental inputs and extending its physical interpretability. While our study focuses specifically on imputing missing heating load data rather than simulating temperature fields, studies from [44,45] that incorporated different physical environmental input values to evaluate model robustness under different conditions provided valuable insights for our study. In our study, these physical inputs represent different weather conditions, resulting in varied model performance depending on the selected variables. We selected widely used weather variables that are known to significantly affect building heating load data into the experiment, such as outdoor temperature, GHI, humidity, and wind speed. We firstly conduct a correlation analysis between heating load data and different weather variables. Table 8 presents the selected weather variables and its symbol using in this paper.

Fig. 25 shows the correlation between the heating load data described in Section 4.1.1 and all weather variables as mentioned in Table 8. The heating load data is highly correlated with the temperature related weather variables, i.e. T_{outdoor} , T_{dew} and T_{ap} , where as the weather variables GHI, φ , v_{wind} has less correlation with heating load data. Other two variables, P and v_{wind} does not show a strong correlation with the heating load data. This further suggests that it is necessary to verify the effectiveness of missing heating load data filling using different physical

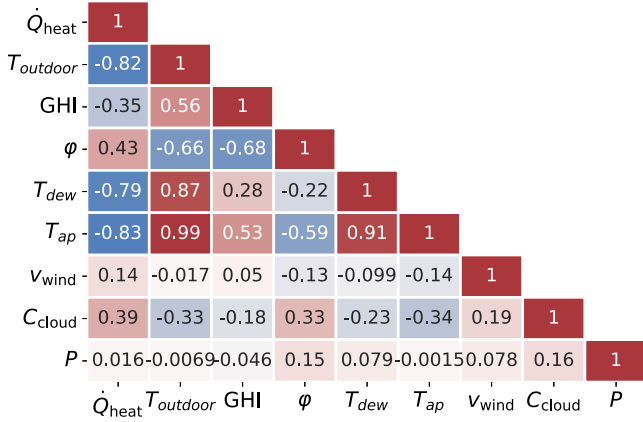


Fig. 25. Pearson correlation between heat load data and different weather variables.

environmental inputs based on our method. Based on this conclusion, we first divide the evaluation of the model performance using different weather variables into two parts: using temperature-related variables and without using temperature related variables.

Fig. 26 shows the comparison of NRMSE when selecting weather variables with and without temperature variables to fill the missing heating load data. The white cube is the average NRMSE. From this demonstration, we can find that the effectiveness of missing heating load data is significantly better when temperature weather variable is used, which lead to a relative reduction of NRMSE for 24.089 % across all used buildings. To better demonstrated the effectiveness when using temperature variables. Fig. 27 shows the difference between using and without using the temperature-related variables. As can be seen from the figure, filling missing values using weather variables unrelated to temperature did not accurately reflect changes in the trend. Furthermore, when filling the missing data during warm months, it also failed to reconstruct data that conformed to the temporal pattern.

One reason why using temperature variables is more effective is that, they are calculated by combining multiple non-temperature variables. T_{ap} is calculated using v_{wind} , φ and T_{outdoor} , while T_{dew} is calculated using φ and T_{outdoor} . Compared to non-temperature variables, they provide information that is more relevant to the changes in heating load data. The differences of correlation are also showed in Fig. 25. Non-temperature variables, i.e. GHI, φ , v_{wind} , P and v_{wind} , have a maximum correlation of 0.43 with heating load data. In contrast, the three temperature-related variables have a minimum absolute correlation of 0.8. The model achieves better performance when weather variables exhibit stronger correlations with heating load, because these highly correlated contextual relationships enable it to learn more meaningful mappings between the heating load data and other variables.

The next step is to evaluate the impact of different temperature variables for the filling of missing heating load data. We compared four different cases. In the first three cases, one temperature-related variable (T_{outdoor} , T_{dew} or T_{ap}) was used alongside a non-temperature variable. In the last case, however, both input weather variables were temperature-related. Fig. 28 shows the comparison of NRMSE when using different temperature variables to fill the missing heating load data. The results indicate that using T_{dew} alongside other weather variables to fill the missing heating load data slightly worsens performance compared to using the other two temperature variables (T_{outdoor} and T_{ap}), especially during the warm months. This causes a relative increase in the NRMSE for 11.164 % and 11.839 %, compared to using T_{outdoor} and T_{ap} , respectively. On the other hand, there was no significant difference for the performance when using only T_{outdoor} or T_{ap} , or when using only temperature-related weather variables.

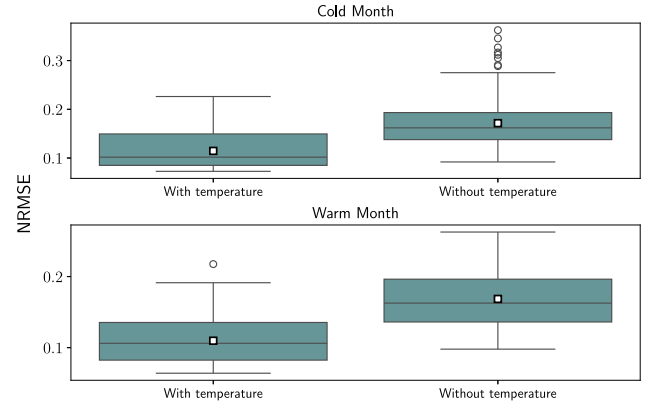


Fig. 26. Comparison of NRMSE between using weather variables with and without temperature-related variables.

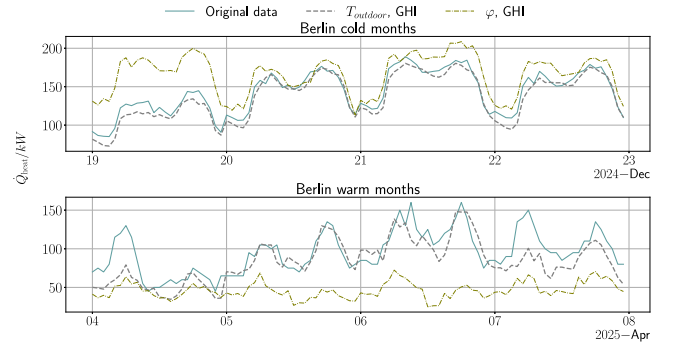


Fig. 27. Correlation between original and imputed data with and without using temperature-related variables.

By comparing the filled data with the original data, it can be seen that the results obtained are consistent with the NRMSE comparison results presented in Fig. 29. Using T_{outdoor} exhibits the better result than using other temperature variables, while using T_{ap} yields nearly identical results. The different results obtained using different temperature-related weather variables can also be explained by the correlation analysis presented in Fig. 25. From the figure, T_{outdoor} and T_{ap} The absolute correlation coefficient between heating load data and the two temperature variables, T_{outdoor} and T_{ap} , are 0.82 and 0.83 in absolute value respectively. This is slightly higher than the absolute correlation coefficient of 0.79 between T_{dew} and heating load data in absolute value. A slightly weaker correlation leads to poorer performance when using T_{dew} , as seen in the deviations from the original data between ‘2023-12-20’ and ‘2023-12-20’ (cold months), as well as between ‘2024-04-05’ and ‘2024-04-07’ (warm months), as shown in Fig. 29. The high correlation between T_{outdoor} and T_{ap} up to 0.99 can also explain the similar results obtained using both variables. Furthermore, using both T_{outdoor} and T_{ap} at the same time did not significantly improve the imputation of missing values, and there was even a significant deviation compared to the original data even when filling missing data for warm months.

5.4. Evaluation the effectiveness of transfer learning

Finally, the application of transfer learning for buildings with limited amount of measured data. Same as the introduction in Section 2.3, We only use data from a single building to train a source model for use with multiple buildings. This simulates the practical case of real-world applications where collecting large amounts of data for multiple buildings is difficult. Different transfer learning methods that

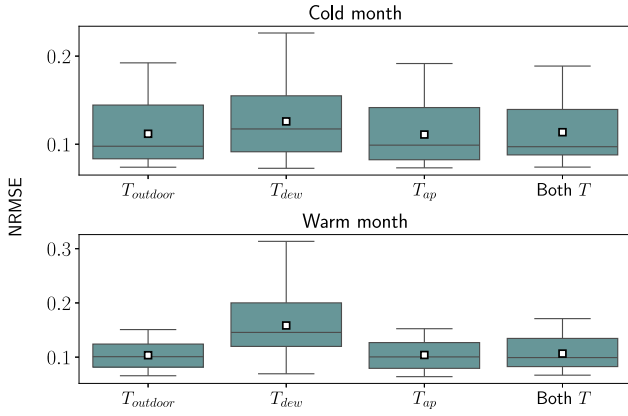


Fig. 28. Comparison of NRMSE using different temperature-related variables.

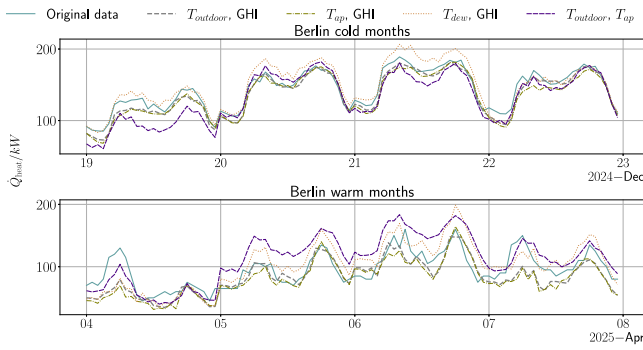


Fig. 29. Correlation between original and imputed data using different temperature-related variables.

have been mentioned in Section 3.3 are then used and compared their performance. We used these methods to fill in long-term missing values for both cold and warm months. The available data for both cases comes from the time range ‘2023-11-10 00:00:00’ to ‘2023-11-30 23:00:00’ and ‘2024-04-10 00:00:00’ to ‘2024-04-30 23:00:00’, respectively. Test data was selected for every target buildings between ‘2024-12-06 00:00:00’ to ‘2025-01-05 23:00:00’ (cold months) and ‘2025-04-01 00:00:00’ to ‘2025-04-30 23:00:00’ (warm months), in order to evaluate the effectiveness between different transfer learning approaches and the effectiveness between with and without using transfer learning methods on training with limited data.

Fig. 30 shows a comparison of the effectiveness of using a transfer learning approach as well as directly training with limited data. The white cube is the average NRMSE. As can be seen from the figure, the reconstruction error obtained through training using transfer learning is lower than that obtained through direct training using limited data. Compared to directly training, using transfer learning approaches achieved a relative reduction of the NRMSE from 3.877 % to 15.847 % in cold months and from 7.488 % to 12.291 % in warm months. When it comes to the comparison between different transfer learning methods, the fine-tuning and domain adaptation method using MMD are better than the instance-based method for filling long-term missing values in both cold and warm months. A more intuitive comparison also showed in Figs. 31 and 32, which visualizes the comparison between the original data and the filled data using different methods. As can be seen, the filled data obtained using instance-based methods and direct training shows a more obvious offset compared to the original data. Furthermore, the reconstructed values for cold months contain more noise, and the correct patterns were not reconstructed for warm months. Fine-tuning and domain adaptation methods, on the other

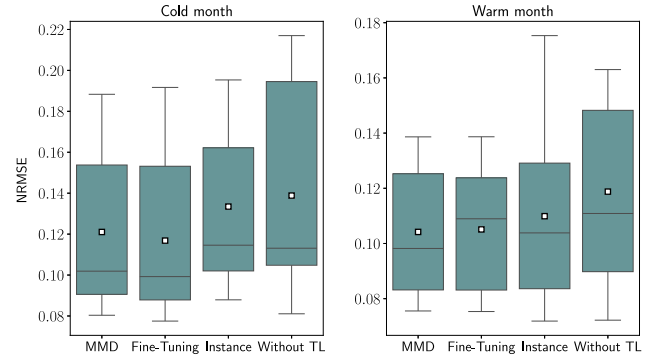


Fig. 30. Comparison of NRMSE between using different transfer learning methods and training from scratch.

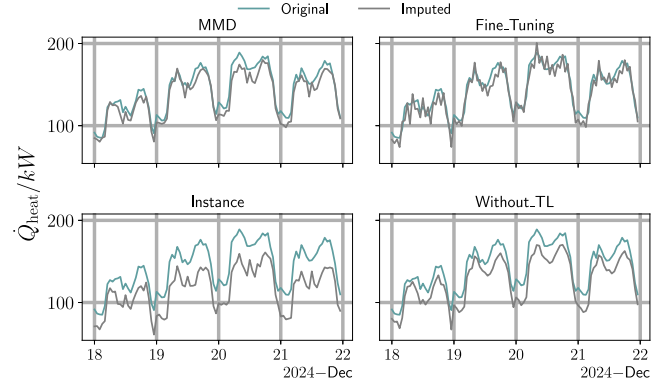


Fig. 31. Comparison between using different transfer learning methods and training from scratch in cold month.

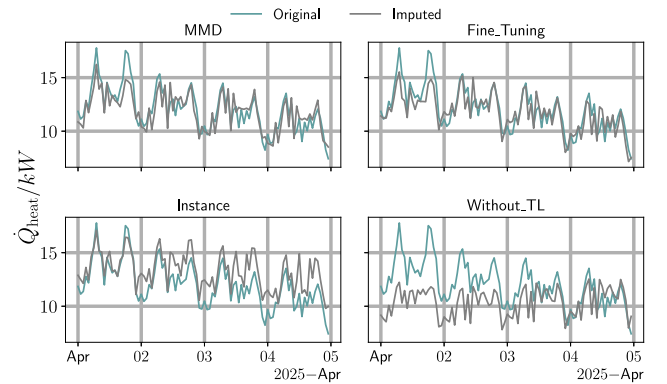


Fig. 32. Comparison between using different transfer learning methods and training from scratch in warm month.

hand, are closer to the original data, which is also consistent with the conclusions drawn from the NRMSE comparison results in Fig. 30.

5.5. Limitations

Despite the fact that our proposed method has been experimentally proven to achieve good results across different seasons and different building types, this work still suffers from the following limitations. Firstly, the performance of the model from warm seasons on the Niš residential datasets is unknown, because all the heating load data in

this data from warm seasons are 0, making the validation on this data meaningless. Besides, due to the lack of additional building-specific metadata such as building insulation, HVAC system type, and occupancy patterns, we cannot determine whether these variables have an impact on the accuracy of estimating the unobserved heating load data using our model. Furthermore, as the When2Heat and Niš datasets contain limited information on various weather variables, the conclusions drawn in Section 5.3 regarding the impact of weather cannot be further validated using these two datasets.

6. Conclusion

In this paper, we proposed a generative network on estimating the unobserved heating load data with weather and temporal data. The experiments were carried out on different heating load datasets, and comparisons with other SOTAs such as Transformer and Informer, in order to prove the effectiveness of our proposed approach. Compared to the baseline models, our proposed method achieved the relative reduction of the NRMSE from 1.65 % to 41.38 %, 0.30 % to 66.43 %, and 14.84 % to 50.22 % on the datasets from three different data sources across different seasons. Another important aspect of this paper is comparing the performance of the model when different weather variables are selected during the filling process using this proposed method. We found that using temperature-related variables significantly improved model performance. In addition, applying different transfer learning methods also showed its ability to fill the long-term missing values when there is only limited available data for a single building, achieving a reduction in NRMSE from 3.88 % to 15.85 % in cold months and from 7.49 % to 12.29 % in warm months compared to not applying the transfer learning methods.

In the future, other more advanced generative networks, such as pix2pix and CycleGAN, will be modified and applied to the task of estimating the unobserved heating load data to see if they perform better. In addition, applying this generative network to multi-task learning will be further explored, so that the network can process the heating load data cross different buildings simultaneously.

CRedit authorship contribution statement

Mengbo Yu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Alexander Neubauer:** Writing – review & editing. **Pedram Babakhani:** Writing – review & editing. **Stefan Brandt:** Project administration. **Martin Kriegel:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Network training

As in the case of the adversarial training network described in [46], in our network, the generator and the discriminator collaborate jointly in the same way: During each training iteration, the energy load data after being filled through the generator will be fed into the discriminator, which will discriminate the generated data by deriving the probability of whether the generated data came from the ground truth. Once the discriminator was trained after each training iteration, the generator will be updated accordingly, in order to maximize the probability from the discriminator and therefore, make the generated results from the generator as close as possible to the ground truth.

The whole training procedure was defined as follow: Let $G(\cdot)$ and $D(\cdot)$ represent the generator and discriminator as the demonstration in Fig. 2 respectively, $G(\cdot)$ was firstly trained for t_G epochs. Once the separate training of $G(\cdot)$ finished, $D(\cdot)$ would then join in the training process and trained with $G(\cdot)$ jointly for t_D epochs. The training procedure was designed in this way, in order to allow the pre-trained generator to produce some meaningful outputs to serve as input to the discriminator, ensuring a stable starting point for the adversarial training process and avoiding the instability and potential mode collapse caused by training with random initialization, which are the common problems encountered with other generative network methods that generate samples from noise. The discriminator, initialized from scratch, started as a randomly initialized network, which allowed the discriminator to dynamically adapt to the outputs of the generator and provided constructive gradients to guide the improvements of the generator after each iteration of the adversarial training process. By starting from scratch, the discriminator avoids any pre-trained bias and focuses entirely on counteracting the outputs of the generator throughout the training process. Together, this setup fosters a competitive adversarial process in which the pre-trained generator and the scratch discriminator iteratively improve each other's performance. In order to validate, whether it is the optimized training procedure in our cases, other two training procedures were also applied as baselines, i.e. training without discriminator and training the generator and discriminator at the same time.

In this framework, we defined a joint reconstruction loss to calculate both the reconstruction loss and the adversarial loss. MSE loss is used to calculate the reconstruction loss, as shown in Eq. (A.1). Assuming $\mathbf{x}$ is that the original multivariate data contains the heat load data, M_G and M_D are two binary matrix used to indicate whether the variant is observed. The binary masks M_G and M_D take the value 1 for the observed weather and time data, while take the value 0 for the unobserved heat load data, indicating that the heat load data would be estimated by the network. $G(\mathbf{x} \odot M_G)$ represent the data that were processed by the generator $G(\cdot)$. By comparing the original data and the data were processed by the generator, the reconstruction loss L_{rec} is calculated.

$$L_{rec} = \|\mathbf{x} - G(\mathbf{x} \odot M_G)\|^2 \quad (A.1)$$

Adversarial loss, on the other hand, is aimed at maximizing the probability of assigning the correct label to both training examples and samples from $G(\cdot)$. $G(\cdot)$ is trained simultaneously to minimize $\log(1 - D(G(\cdot)))$. The adversarial loss can be defined as:

$$\min_G \max_D \mathbb{E}[\log D(\mathbf{x}, M_D) + \log(1 - D(G(\mathbf{x} \odot M_G), M_G))] \quad (A.2)$$

By combining it with reconstruction loss, the loss function for the whole generative framework can be written as:

$$\min_G \max_D \mathbb{E}[L_{rec}(\mathbf{x}, M_G) + \lambda[\log D(\mathbf{x}, M_D) + \log(1 - D(G(\mathbf{x} \odot M_G), M_G))]] \quad (A.3)$$

where $L_{rec}(\mathbf{x}, \mathbf{M})$ is the reconstruction loss that has been represented in Eq. (A.1), λ is the weighted hyper-parameter. The joint loss is then updated by Eq. (A.4) according to Eq. (A.3).

$$\mathbb{E}[\nabla \theta_D L_{rec}(\mathbf{x}, M_G) + \nabla \theta_D \lambda[\log D(\mathbf{x}, M_D) + \log(1 - D(G(\mathbf{x} \odot M_G), M_G))]] \quad (A.4)$$

Table B.9

Structure of generator in proposed model.

Type	Kernel size	Dilation rate	Stride	Filter
Conv2d.	3	1	2	128
Conv2d.	3	1	2	256
Dilated conv.	3	(1,1)	1	256
Dilated conv.	3	(2,2)	1	256
Dilated conv.	3	(4,4)	1	256
Dilated conv.	3	(8,8)	1	256
Deconv2d.	3	1	1/2	128
Deconv2d.	3	1	1/2	1

Table B.10

Structure of global and local discriminator.

Type	Kernel size	Stride	Filter	Node
Deconv2d.	3	1/2	128	–
Deconv2d.	3	1/2	256	–
FC.	–	–	–	1024

Table B.11

Structure of discriminator after concatenating.

Type	Node
Concat.	2048
FC.	1

Table B.12

Hyper-parameters using in grid search.

Hyperparameter	Values
Learning rate	0.001, 0.0005, 0.00025, 0.0002, 0.0001
Batch size	4, 8, 16, 32, 64

Table B.13

Hyper-parameters using in grid search.

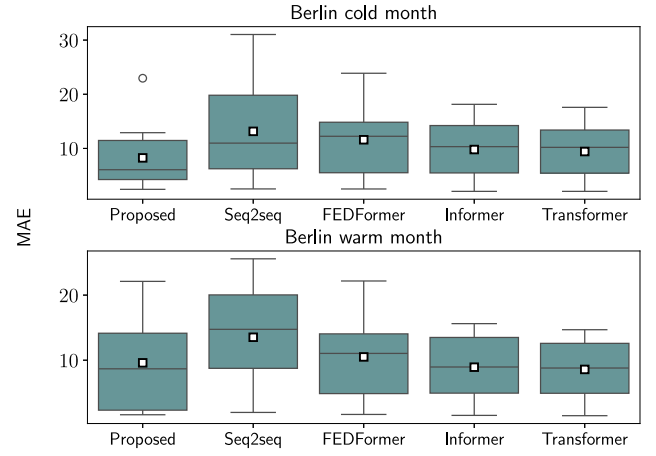
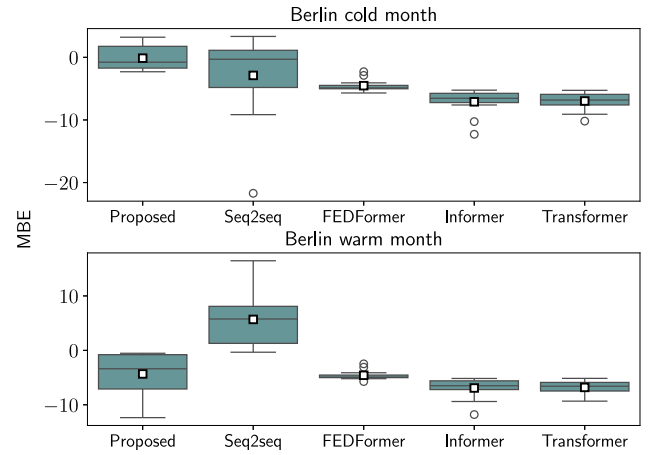
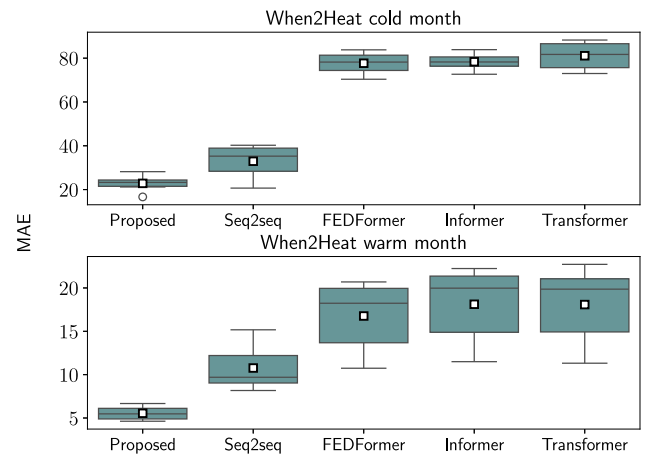
Data source	Learning rate	Batch size	Fixed seed
Berlin residential buildings	0.0001	32	1
When2Heat profiles	0.0001	64	1
Niš DHS infrastructure	0.0001	16	1

Appendix B. Network setup

Table B.9 shows the structure of the generator network. Both the global and local discriminator has the same structure. Tables B.10 and B.11 display the structure of global discriminator and local discriminator respectively, as well as the structure after they are concatenated together. Following the description in Section 3.3, λ for calculating network loss in Eq. (A.3) is set to 0.0004, T_G is set to 100 iterations and T_D is set to 50 iterations. The model optimizer is chosen as Adam. The other two important hyper-parameters, i.e., learning rate and batch size, are determined by the grid search method to ensure that the hyperparameter settings are optimal when training for different buildings. The hyper-parameters chosen by grid search are shown in Table B.12. The exact hyper-parameter setup for the datasets from each data source are shown in Table B.13.

Appendix C. Comparison of MAE and MBE between our proposed method and autoregressive baselines

The results obtained using autoregressive baselines were also compared using other two metrics, MAE and MBE, as shown from Figs. C.33 to C.38. The results demonstrate that our proposed method

**Fig. C.33.** Comparison of MAE on Berlin residential building dataset.**Fig. C.34.** Comparison of MBE on Berlin residential building dataset.**Fig. C.35.** Comparison of MAE on 'When2Heat' dataset.

consistently outperforms other baseline methods in terms of both data accuracy and model bias across a range of datasets. Only when filling in long-term missing values in the Niš dataset, Seq2seq exhibits a smaller bias.

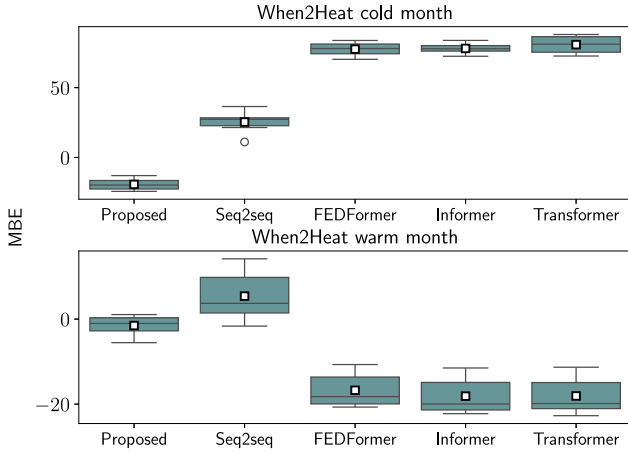


Fig. C.36. Comparison of MBE on 'When2Heat' dataset.

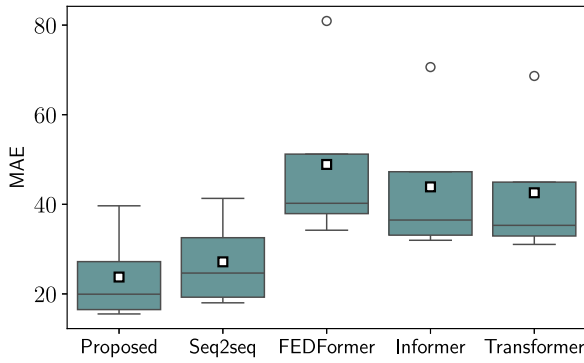


Fig. C.37. Comparison of MAE on Niš dataset.

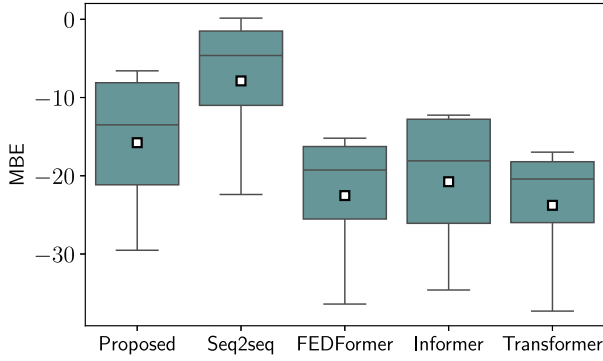


Fig. C.38. Comparison of MBE on Niš dataset.

Appendix D. Comparison of MAE and MBE between our proposed method and non-autoregressive baseline methods with weather-only input

The comparisons of the two other metrics, MAE and MBE, between our proposed method and the two other non-autoregressive baselines (CNN-AE and TCN-AE) are shown from Figs. D.39 to D.44. The results demonstrate the same conclusion as in Section 5.2.2. Although the differences in performance between the three methods were less pronounced on the Berlin residential dataset, our proposed method outperformed the non-autoregressive baseline methods on the other two datasets, especially the Niš dataset (see Figs. D.40–D.43).

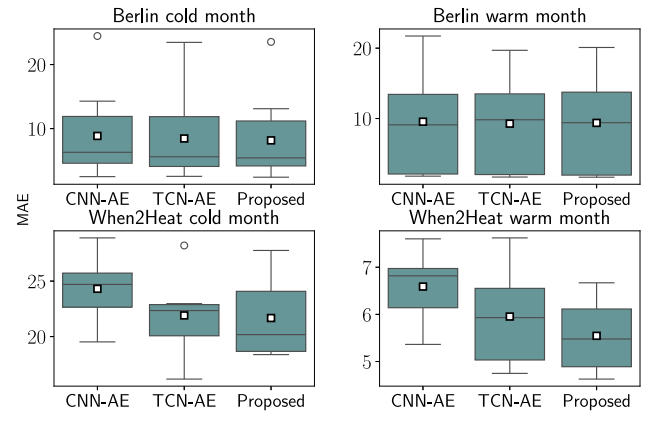


Fig. D.39. Comparison of MAE on Berlin residential building dataset.

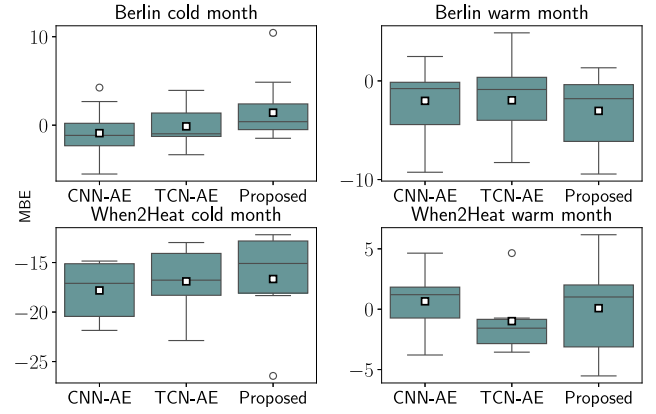


Fig. D.40. Comparison of MBE on Berlin residential building dataset.

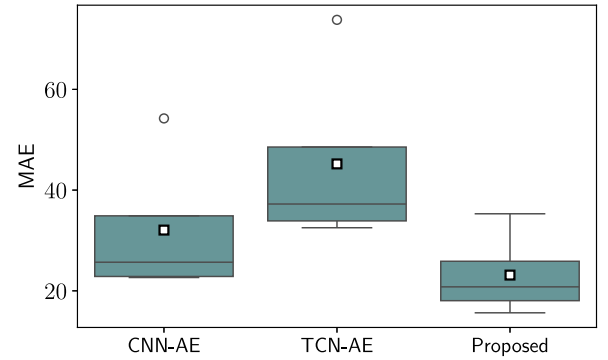


Fig. D.41. Comparison of MAE on 'When2Heat' dataset.

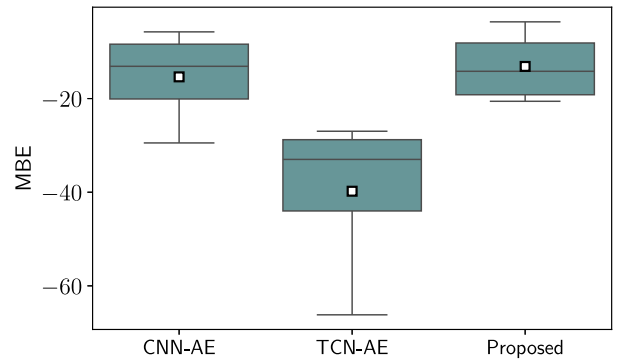


Fig. D.42. Comparison of MBE on 'When2Heat' dataset.

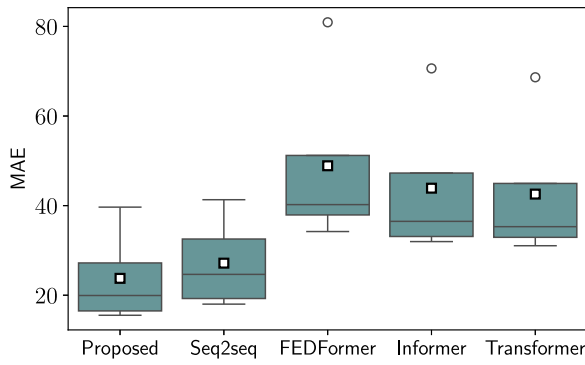


Fig. D.43. Comparison of MAE on Niš dataset.

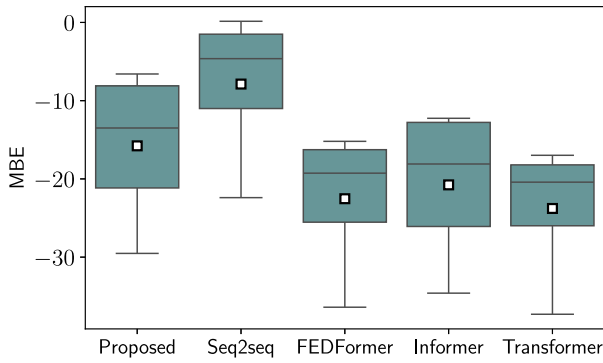


Fig. D.44. Comparison of MBE on Niš dataset.

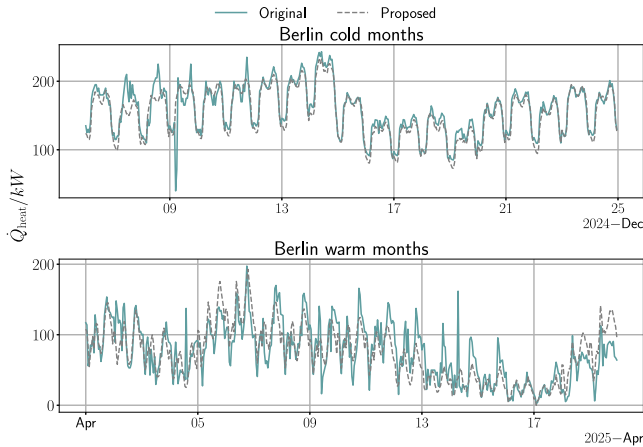


Fig. E.45. Comparison between the original data and imputed data for the Berlin residential dataset across different seasons.

Appendix E. Demonstration of the results using our proposed network

In order to demonstrate the result from our proposed method better, Figs. E.45 to E.47 show the comparison of the original data and imputed data using our proposed method.

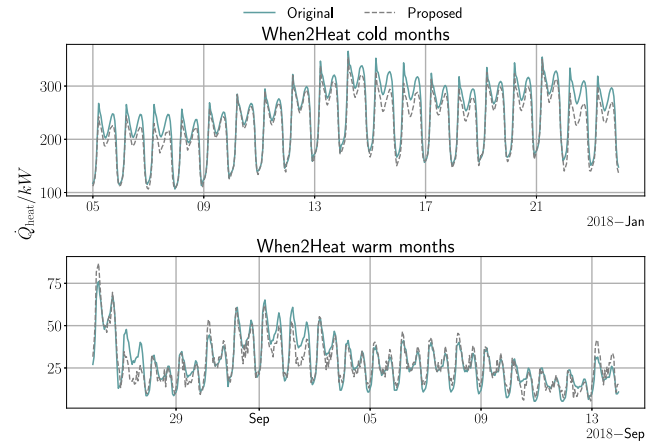


Fig. E.46. Comparison between the original data and imputed data for When2Heat dataset across different seasons.

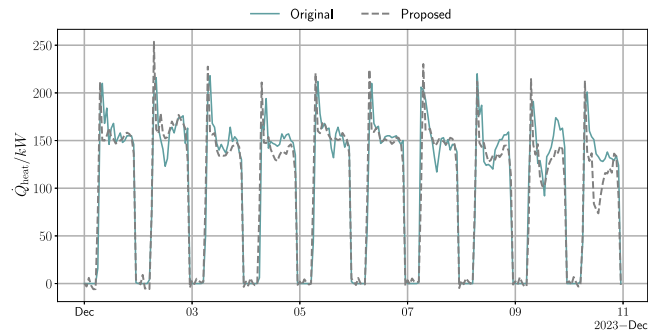


Fig. E.47. Comparison between the original data and imputed data for the Berlin residential dataset across different seasons.

Data availability

The code and data from When2Heat Profile as well as Niš datasets are freely available on https://github.com/you9731/Imputing_network.

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